

M3-Score: A Multi-Axis Medical-MMD Metric with A-Priori Layers and Native Significance Testing for Generative Radiology Image Models

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Abstract—Evaluating generative models for radiology image synthesis is harder than evaluating natural-image generators: the clinically relevant signal lies in rare, high-frequency pathological structure that natural-image embeddings under-represent. We introduce the Medical Multi-axis Maximum Mean Discrepancy (M3-Score), an evaluation framework grounded in the frozen RadiDINO-s16 vision transformer (ViT-S/16) pre-trained on radiology images. Rather than collapsing generation quality into one scalar, M3 reports two separately defined quantities, each read from a transformer layer fixed *a priori* by representational depth and never tuned on test outcomes: a *fidelity* axis (unbiased multi-bandwidth RBF MMD² at the final layer) and a *memorization* axis (nearest-neighbour distances at 75% depth). Each fidelity estimate is reported with a permutation-test *p*-value and a bootstrap 95% confidence interval. On BraTS brain MRI the fidelity axis cleanly separates real from DDPM-generated images (permutation $\hat{p} = 1/501 = 0.002$, the resolution floor at 500 shuffles; observed MMD² = 0.153, bootstrap 95% CI [0.146, 0.162]) and detects out-of-distribution radiology with ROC-AUC = 0.976 versus 0.809 for the InceptionV3 (FID) backbone and 0.659 for CLIP (CMMD). Our central finding is a *structural failure of CMMD*: it *rank-inverts* radiology OOD, assigning a cross-modality chest/lung CT shift a *smaller* distance from brain MRI (0.699, 95% CI [0.686, 0.715]) than a weak same-modality generator (0.738, [0.723, 0.758]; paired $\Delta = -0.039$, $[-0.059, -0.021]$, excluding zero), whereas the M3 fidelity axis orders them correctly (0.248 vs. 0.457, $\Delta = +0.210$). FID (InceptionV3) also orders them correctly, so the failure is specific to CLIP features, not to natural-image backbones in general. Using real expert glioma masks with an area-matched, texture-controlled design, M3 is lesion-selective under subtle degradation (1.4–1.8 $\times$ more responsive to tumour-region than to matched healthy-region perturbation) where an Inception-MMD baseline is not ($\sim 1.05\times$), an effect attributable to the radiology backbone. We also report results that did not favor M3: FID is *more* stable than M3 at small sample sizes ($N < 100$); a third, exploratory coverage diagnostic fails its own mode-drop validation at *every* one of the twelve encoder depths, so it is excluded from all claims here and reported only as a negative result; and the a-priori depth for the memorization axis, while harmless, is not empirically corroborated. M3 is scoped to the radiology (CT/MRI) domain in which its backbone was trained.

Index Terms—Generative models, medical image synthesis, evaluation metrics, maximum mean discrepancy, RadiDINO, interpretability, diffusion models.

I. INTRODUCTION

Generative models are increasingly utilized to address data scarcity of annotated medical imaging data, which continues

to constrain clinical artificial intelligence (AI) development across modalities [1], [2]. Strict patient privacy regulations, the high cost of expert annotation, severe class imbalance due to disease rarity, and logistical challenges in multi-center data sharing collectively limit the creation of large-scale, high-quality datasets [3], [4]. Following the pioneering role of Generative Adversarial Networks (GANs) [5] in medical image synthesis, Denoising Diffusion Probabilistic Models (DDPMs) [6] have substantially elevated the state of the art, enabling the generation of highly realistic and anatomically consistent images across MRI, CT, chest radiography, retinal fundus photography, dermoscopy, and ultrasound [2], [7], [8]. Downstream benefits, including improvements to segmentation, classification, and reconstruction performance through synthetic data augmentation, have been demonstrated across these modalities, particularly for rare-pathology cohorts [1], [3], [4].

Yet, deploying generative models for medical image synthesis in clinical or research pipelines imposes a stringent validation burden that far exceeds the capabilities of conventional natural-image benchmarks. A synthetically generated medical image may attain excellent scores on standard quantitative metrics while still containing anatomically implausible structures, globally inconsistent intensity distributions, or clinically hazardous hallucinations [9], [10]. These artifacts are immediately evident to domain experts yet remain undetectable by conventional evaluation measures. This critical mismatch constitutes the central motivation for the present work.

A. Limitations of Existing Evaluation Metrics

The Fréchet Inception Distance (FID) [11] and Kernel Inception Distance (KID) [12] are among the most widely used evaluation metrics for generative image models. Both compute distributional distances in a feature space extracted by InceptionV3 [13], a classifier pre-trained on ImageNet. This choice of feature extractor creates three structural failure modes when applied to medical image evaluation.

a) *Domain mismatch*: InceptionV3 was trained to classify natural images into everyday categories such as animals, objects, and scenes, and its internal representations are therefore tuned to natural-image statistics. Consequently, it is insensitive to clinically relevant features in medical images, for example tissue texture heterogeneity in MRI, Hounsfield unit distributions in CT, vascular branching patterns in retinal fundus photography, and ground-glass opacity patterns in chest

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Manuscript received August 31, 2026.

radiography [9], [14]. Kelkar *et al.* [9] showed that state-of-the-art GANs can obtain favorable FID scores on medical images while failing to preserve per-image statistics that are meaningful for downstream clinical tasks. Recent work by Woodland *et al.* [15], evaluating sixteen generative models across four medical imaging modalities, found that ImageNet-based feature extractors can be more consistent with human expert judgment than single-modality medical encoders. This highlights that the relationship between backbone domain specificity and metric reliability is non-trivial, motivating the need for a purpose-built multi-modality radiology encoder.

b) Aggregation over heterogeneous populations: FID and KID reduce the distribution of generated images to a single scalar, which can hide heterogeneous structure in the generated sample space. In our brain MRI experiments, a DDPM attains a competitive FID (68.8) while its generated samples occupy near-zero manifold precision relative to the real data (α -Precision ≈ 0 ; Section IV-C). A favorable FID therefore does not imply that the generator covers the real distribution, and a metric that separates these failure modes is needed to tell the two cases apart.

c) Absence of per-image and per-scale interpretability: Neither FID, KID, nor manifold-based metrics [16], [17] provide per-image scores or a decomposition of distributional shift across clinical domains. A practitioner cannot determine from a single-scalar FID value whether a generated dataset suffers from global intensity drift, geometric distortion, or semantic incoherence, nor identify which individual images are safe to use downstream. This interpretability gap represents a fundamental obstacle to responsible deployment of generative models in medical imaging pipelines.

Pixel-level metrics such as Structural Similarity Index (SSIM) [18] and Peak signal-to-noise ratio (PSNR) are widely used in medical image synthesis evaluation [19], yet both carry well-documented limitations. SSIM is insensitive to blurring, unreliable under intensity shifts in unnormalized images, and shows poor correlation with human radiological perception for images containing subtle pathological findings [19], [20]. PSNR is highly sensitive to normalization choices and inconsistent across different intensity ranges, and prior work recommends against its use in medical image quality assessment [19]. More broadly, Deo *et al.* [10] demonstrated that nearly all standard upstream metrics are insensitive to localised, clinically relevant morphological alterations such as distorted tumour boundaries, and that metric rankings of generative models do not align with downstream segmentation performance. Manifold-based metrics [16], [17] partially address distributional diversity collapse but carry substantial computational overhead that limits their use in training-time monitoring workflows.

B. The M3-Score: Design Rationale

We propose the **Medical Multi-axis Maximum Mean Discrepancy (M3-Score)**, an interpretable evaluation framework for generative radiology image models. The M3-Score rests on three design decisions, each motivated by the limitations identified above.

a) Domain-specific feature extraction via RadioDINO:

Rather than relying on InceptionV3 features calibrated to natural-image statistics, the M3-Score extracts frozen Vision Transformer (ViT) embeddings from RadioDINO [25], a self-supervised encoder pre-trained on radiology images via DINOv2-style distillation [26]. Because RadioDINO encodes textural and structural invariances that are clinically relevant in radiology, including tissue boundary sharpness, organ morphology regularity, and pathological texture heterogeneity, the M3-Score is applicable across radiology modalities (CT, MRI) without backbone retraining. We use the compact **RadioDINO-s16** variant (ViT-S/16, 12 blocks, 384-d embeddings); the backbone comparison (Section IV-B) shows it attains the highest OOD ROC-AUC (0.976 at $N=500$; 0.995 at $N=100$) among six candidate backbones, ahead of the larger Microsoft rad-dino ViT-B/14 (0.913) and generic DINOv2 (0.968). We deliberately restrict claims to the radiology domain: features for retinal fundus or other non-radiology modalities fall outside the backbone's training distribution and M3 values there are not directly comparable (Section IV-B).

b) Three a-priori axes instead of one tuned scalar:

Distinct generation failures manifest at distinct representational depths, so M3 reports three separately defined quantities rather than a single weighted scalar. Critically, each axis reads from a layer fixed *before any test-set result is observed*, by representational position alone: the **fidelity** axis is an unbiased multi-bandwidth RBF MMD² [27] at the final block (L12, the deepest semantic representation, analogous to FID's use of the InceptionV3 penultimate activation); the **memorization** axis uses nearest-neighbour distances at 75% depth (L9), where features retain instance-level variation that the final layer collapses; the **coverage** axis uses k-NN manifold precision/recall [16] at 33% depth (L4), where structural diversity is encoded. No layer, threshold, or kernel bandwidth in M3 is selected by which choice makes a reported result look best; a post-hoc CKA ablation (Section IV-H) independently confirms that the final layer maximises real-versus-generated discriminability for the fidelity axis, but the axis assignment does not depend on that ablation.

c) Native statistical reporting: For the fidelity axis, M3 reports a permutation-test p -value and a bootstrap 95% confidence interval. FID, KID, and CMMD provide neither. On $N = 500$ images the fidelity evaluation runs in approximately 8 s on a single consumer GPU, making it viable as a checkpoint monitor during training.

C. Contributions

The contributions of this paper are as follows.

1. **Three-axis M3 formulation with a-priori layers.** A radiology-domain evaluation framework that reports fidelity (final-layer unbiased RBF MMD²), memorization (mid-deep nearest-neighbour distances), and coverage (early-mid k-NN precision/recall) as three separately defined quantities. Every layer is fixed by representational position before any test result is seen, eliminating the selection-on-outcome bias that inflates single-scalar metrics.

TABLE I: Unified benchmark of evaluation metrics for generative medical image models. The M3-Score reports two validated axes read from layers fixed *a priori* by representational depth (fidelity L12, memorization L9); the fidelity axis additionally carries a native permutation *p*-value and bootstrap CI (Section I-B). A third, exploratory coverage diagnostic at L4 is defined in Section III-B but excluded from this comparison and from every claim in this paper: it fails its own validation at all twelve encoder depths (Sections IV-C and IV-D).

Category	Metric	Dir.	Range	Description
Distributional	FID [11]	↓	$[0, \infty)$	Fréchet distance between InceptionV3 feature distributions of real and generated images.
	KID [12]	↓	$[0, \infty)$	Unbiased kernel-based distributional distance; more stable than FID at small sample sizes.
Perceptual	SSIM [18]	↑	$[0, 1]$	Structural similarity combining luminance, contrast, and structure terms.
	PSNR	↑	$[0, \infty)$	Peak signal-to-noise ratio; sensitive to normalization choices.
	MS-SSIM	↑	$[0, 1]$	Multi-scale extension of SSIM for perceptual consistency at varying resolutions.
	LPIPS [21]	↓	$[0, \infty)$	Learned perceptual image patch similarity using deep visual embeddings.
	IS [22]	↑	$(0, \infty)$	Inception Score; measures sample diversity and classifier confidence.
Manifold	α -Precision [16], [23]	↑	$[0, 1]$	Fraction of generated samples lying within the real data manifold (fidelity).
	β -Recall [16], [23]	↑	$[0, 1]$	Fraction of real manifold covered by generated samples (diversity).
	Authenticity	↑	$[0, 1]$	Evaluates whether generated samples avoid memorization of training images.
	Coverage [24]	↑	$[0, 1]$	Fraction of real samples represented by at least one generated neighbour.
	Density [24]	↑	$[0, \infty)$	Local concentration of generated samples near the real data manifold.
M3-Score (ours)	M3 Fidelity Axis	↓	$[0, \infty)$	Unbiased multi-bandwidth RBF MMD ² between RadioDINO-s16 final-layer (L12) features of real and generated images, with a native permutation <i>p</i> -value, bootstrap CI, and null-standardized separation <i>Z</i> .
	M3 Memorization Axis	↓	$[0, 1]$	Fraction of generated images whose nearest real neighbour at L9 is closer than the median within-real neighbour gap; flags near-duplicate leakage.
	M3 Coverage Diagnostic (<i>Experimental</i>)	↑	$[0, 1]$	<i>k</i> -NN manifold precision/recall at L4. <i>Experimental: unstable under k-NN radius inflation at small N; see Section IV-C. Not a validated axis.</i>

- Native statistical inference for the fidelity axis.** Each fidelity estimate is accompanied by a permutation-test *p*-value and a bootstrap 95% confidence interval, neither of which FID, KID, or CMMD provide. We additionally report a null-standardized separation *Z*, but only as a descriptive signal-to-noise ratio: it grows with *N* by construction and carries no inferential content (Section IV-I).
- Identification of a CMMD structural failure.** We show that CMMD (CLIP ViT-L/14 features) *rank-inverts* a weak same-modality MRI generator and a full cross-modality CT shift, ranking the CT as closer to real MRI (paired $\Delta = -0.039$, 95% CI $[-0.059, -0.021]$, excluding zero), whereas the M3 fidelity axis, and InceptionV3/FID, order them correctly. This is, to our knowledge, the first documented rank-inversion failure of CMMD on a medical OOD ordering task.
- Per-axis validation, including negative results.** We vali-

date the memorization axis by injecting near-duplicates at known rates (recovered exactly) and probe the coverage axis with mode-drop. Three results run against the metric and are reported in full: (i) FID is more sample-stable than M3 at $N < 100$, (ii) the coverage axis is unstable under *k*-NN radius inflation, and (iii) using *real* expert lesion masks, M3 shows no lesion-specific advantage under gross erasure, though it is lesion-selective ($1.4\text{--}1.8\times$ vs. an Inception-MMD baseline's $\sim 1.05\times$) under subtle noise, a backbone-attributable *specificity* to subtle diagnostic-region degradation rather than the larger raw-sensitivity figure claimed in early drafts.

- Open, protocol-harmonized pipeline.** A single canonical configuration (RadioDINO-s16, $N = 500$, multi-bandwidth RBF, U-statistic MMD²) is applied across every experiment, with explicit acceptance criteria and released code and layer caches.

D. Scope

The primary validation domain is axial brain MRI slices from the BraTS dataset [28], [29], chosen for its established benchmark status, large corpus size (38,781 slices), and multi-pathology diversity. Cross-domain experiments additionally use chest/lung CT (LIDC-style) and a non-radiology modality (retinal fundus) to probe the limits of the backbone.

We explicitly scope M3 to the **radiology (CT/MRI) domain** in which RadioDINO was trained. This is a deliberate restriction, not an under-statement: our cross-modality experiment (Section IV-B) shows that M3 ranks retinal fundus images as *further* from brain MRI than chest CT is, which is correct under a radiology-domain ontology ($CT \approx MRI \gg \text{fundus}$) but means raw M3 values are not comparable across the radiology boundary. Applying M3 to a new *radiology* modality requires only a paired real-vs-generated set and the frozen encoder; applying it outside radiology would require a different backbone.

The M3-Score complements rather than replaces FID [11] and SSIM [18]: reporting M3 alongside these standard metrics adds a domain-calibrated fidelity axis with native significance testing plus memorization and coverage diagnostics. Method limitations, including the cases where M3 does not outperform existing metrics, are discussed in Section V.

E. Paper Organization

Section II reviews related work on generative medical image synthesis and evaluation metrics. Section III describes the methodology, including the canonical three-axis a-priori formulation and the legacy post-hoc validation machinery. Section IV reports experimental results, including those that do not favor M3. Section V discusses findings, limitations, and the radiology-domain scope. Section VI concludes.

II. RELATED WORK

Evaluating generative image models is a long-standing open problem. The field has evolved through three broad phases: hand-crafted pixel-level metrics, learned distributional metrics anchored to natural-image classifiers, and, most recently, task-aware or domain-specific metrics motivated by the inadequacy of general-purpose feature extractors for medical imaging. This section reviews the principal contributions in each phase and positions the M3-Score within this trajectory.

A. Phase 1: Pixel-Level and Structural Metrics

The earliest quantitative metrics for image quality operate at the pixel level. PSNR measures reconstruction fidelity as the logarithm of the ratio between the maximum possible signal and the mean squared error, making it sensitive to normalization choices and inconsistent across intensity ranges. Wang *et al.* [18] introduced the SSIM, which partially addresses these limitations by jointly evaluating luminance, contrast, and structural consistency within local windows. SSIM became the de facto paired-image metric in medical image synthesis [19] and remains widely reported alongside distributional metrics. However, Bréger *et al.* [20] and Dohmen *et al.* [19] have

both shown that SSIM correlates poorly with radiological perception for images containing subtle pathological findings, and that PSNR is unreliable under the normalization conventions common in medical imaging. Zhang *et al.* [21] introduced the Learned Perceptual Image Patch Similarity (LPIPS) metric, which replaces hand-crafted structural terms with a calibrated distance in a pretrained deep feature space. While LPIPS aligns better with human judgement than PSNR or SSIM on natural images, it inherits the ImageNet bias of its backbone and is designed for paired comparisons, limiting its applicability to unpaired generative model evaluation.

B. Phase 2: Distributional Metrics and Modern Embeddings

1) Distributional Metrics Based on Inception Features:

The dominant paradigm for evaluating unpaired generative models originated with the Inception Score (IS) introduced by Salimans *et al.* [22], which measures the sharpness and diversity of generated images by applying a pretrained InceptionV3 classifier [13] to the generated set and computing the KL divergence between the conditional and marginal label distributions. IS is computed on generated images alone, with no reference to real data, and therefore cannot detect mode collapse or distribution shift.

FID, introduced by Heusel *et al.* [11] as part of a two-time-scale GAN training framework, addressed this limitation by computing the Fréchet distance between multivariate Gaussians fitted to the InceptionV3 feature distributions of real and generated images. FID rapidly became the primary benchmark metric across the generative modeling literature and remains the most reported scalar in medical image synthesis evaluations. However, FID carries three structural limitations that are particularly acute in the medical setting: it assumes Gaussian-distributed activations (which is violated by ReLU features), it is a biased estimator whose bias scales inversely with sample size, and its backbone encodes natural-image rather than clinical image statistics [9], [14].

KID, proposed by Binkowski *et al.* [12], addressed the bias and distributional assumptions of FID by replacing the Fréchet distance with an unbiased Maximum Mean Discrepancy (MMD) estimator using a polynomial kernel over the same InceptionV3 features. KID admits an analytic unbiased estimator, is more reliable than FID at the small sample sizes typical in medical imaging, and does not presuppose a parametric distributional form. The underlying MMD framework was formalised by Gretton *et al.* [27] as a nonparametric two-sample test whose statistic is the largest difference in expectations over functions in a reproducing kernel Hilbert space. KID retains the InceptionV3 backbone and therefore shares the domain mismatch problem of FID, though its statistical properties are considerably stronger.

2) Moving Beyond Inception: Richer Embeddings and Kernels:

Jayasumana *et al.* [30] proposed CLIP Maximum Mean Discrepancy (CMMD) as a direct replacement for FID. CMMD computes an unbiased Gaussian RBF kernel MMD over CLIP [31] embeddings, which encode richer semantic content than InceptionV3 because CLIP is trained on diverse image-text pairs at scale. The authors demonstrated

empirically that FID contradicts human raters, fails to reflect gradual quality improvement in iterative text-to-image models, and can paradoxically improve when progressive distortions are applied; CMMD behaves monotonically and correctly under all three conditions. CMMD is distribution-free, sample-efficient, and provides an unbiased estimate. However, CLIP embeddings are still derived from natural image-text pairs and do not encode the anatomical and pathological structure of medical images.

C. Phase 3: Manifold-Based, Domain-Specific, and Radiology-Aware Metrics

1) *Manifold-Based Fidelity and Diversity Metrics*: A complementary line of work decouples the fidelity and diversity aspects of generative model quality that FID collapses into a single scalar. Kynkäänniemi *et al.* [16] introduced improved precision and recall metrics that form explicit, non-parametric approximations to the real and generated data manifolds using k -nearest neighbour sets: Precision measures the fraction of generated samples that fall within the real manifold, and Recall measures the fraction of the real manifold covered by generated samples. Naeem *et al.* [24] subsequently showed that the Kynkäänniemi precision and recall metrics are unreliable in the presence of outliers and fail to detect identical distributions; they proposed Density and Coverage as more robust alternatives that use overlapping manifold hyperspheres rather than strict membership. These manifold metrics provide complementary diagnostic information but require dense sampling to form reliable manifold estimates, making them computationally prohibitive at training-time monitoring scales.

2) *Domain-Specific Metrics for Medical Image Evaluation*: The inadequacy of ImageNet-pretrained feature extractors for medical image evaluation has motivated a small but growing body of domain-specific metrics. Kelkar *et al.* [9] demonstrated that state-of-the-art GANs can attain favorable FID scores on medical images while failing to preserve per-image statistics that matter for downstream clinical tasks. Deo *et al.* [10] showed that nearly all upstream metrics are insensitive to localised morphological distortions such as tumour boundary deformation, and that metric rankings do not align with downstream segmentation performance.

Konz *et al.* [32] introduced the Fréchet Radiomic Distance (FRD), the most direct precursor to the M3-Score in the medical imaging literature. FRD replaces deep learned features with standardised radiomic features from PyRadiomics, computing the Fréchet distance between Gaussian-fitted radiomic distributions of real and generated images. FRD was shown to outperform FID, KID, and CMMD across OOD detection, image-to-image translation evaluation, and unconditional generation quality, with additional advantages of stability at small sample sizes and correlation with radiologist-perceived quality; it has been published in Medical Image Analysis [32].

The M3-Score differs from FRD in three respects. First, it uses deep learned features from a radiology-pretrained ViT (RadioDINO [25]) rather than hand-crafted radiomic features, capturing higher-order structural and semantic representations not available to first-order or texture-based descriptors. Sec-

ond, rather than a single scalar, M3 reports three separately-defined axes read from *fixed a-priori* layers (fidelity at L12, memorization at L9, and coverage at L4), which decouple distinct failure modes. Third, the fidelity axis is an *unbiased* multi-bandwidth RBF MMD² reported with a native permutation p -value and bootstrap CI, whereas FRD computes a Fréchet (Gaussian-moment) distance without a built-in significance test. We were unable to include FRD in our experimental comparison because its PyRadiomics Laplacian-of-Gaussian filter is incompatible with our BraTS axial-slice dimensions (Section IV-A); we note this as a limitation of our evaluation, not of FRD, whose reported strengths on medical tasks make it a valuable complementary metric.

3) *Self-Supervised Radiology Encoders*: The M3-Score's feature extractor is RadioDINO, introduced by Pérez-García *et al.* [25], a family of Vision Transformer encoders pre-trained on over 838,000 radiology images spanning multiple modalities and anatomical regions using the DINOv2 [26] self-supervised distillation objective; we adopt the compact ViT-S/16 variant (RadioDINO-s16). RadioDINO was trained without text supervision, using image-level and patch-level objectives, and was shown to match or exceed the performance of text-supervised biomedical encoders on classification, segmentation, and report generation benchmarks. The absence of text supervision and the breadth of the radiology training corpus make RadioDINO's intermediate ViT representations more likely to encode the anatomical and pathological structure relevant to generation quality assessment than either CLIP or InceptionV3 features [25].

Woodland *et al.* [15] conducted a large-scale study evaluating sixteen generative models across four medical imaging modalities and found that the choice of backbone substantially affects metric consistency with human expert judgment, motivating the use of a purpose-built multi-modality radiology encoder rather than a single-modality or general-purpose backbone.

Three recurring gaps emerge from the work reviewed above: (a) dependence on general-purpose encoders misaligned with radiology semantics, (b) collapse of evaluation into a single scalar that conflates distinct failure modes (fidelity, memorization, coverage), and (c) absence of native significance testing and per-image interpretability. M3-Score addresses all three by building on a purpose-built radiology encoder (RadioDINO-s16), reporting separate fidelity and memorization axes read from layers fixed *a priori* by representational depth, and accompanying the fidelity axis with a permutation p -value and a bootstrap confidence interval. Section III develops the construction.

III. METHODOLOGY

This section presents the M3-Score in full technical detail. Section III-A introduces the notation and the canonical pipeline (Algorithm 1). Section III-B defines the three *a-priori* axes. Section III-C defines the feature-extraction procedure and Section III-D the per-layer unbiased MMD² estimator. Section III-E defines the per-image score, Section III-F the computational complexity, and Section III-G the procedure

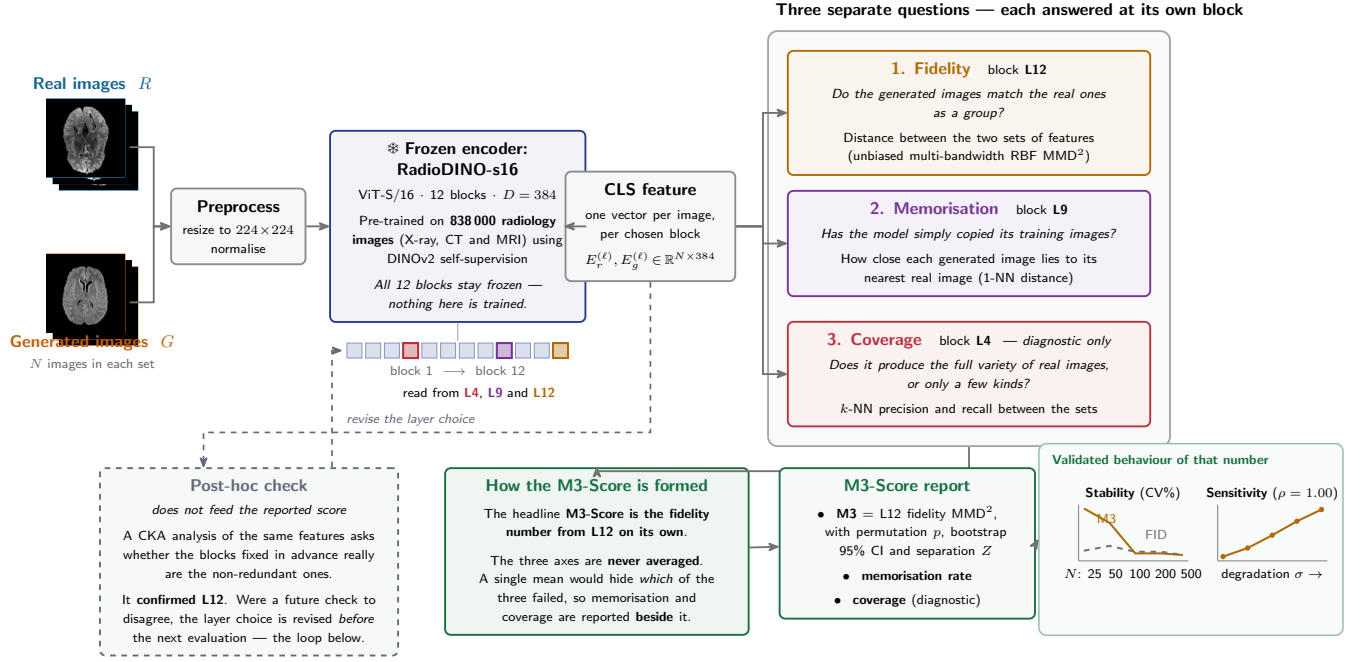


Fig. 1: Overview of the M3-Score evaluation pipeline. The real set R and generated set G (N images each) pass through a single frozen RadioDINO-s16 encoder (ViT-S/16, pre-trained on 838,000 radiology images with DINOv2 self-supervision). No part of the encoder is trained or fine-tuned: this is an evaluation-time pipeline only. One CLS-token feature vector per image is read from each of three *a-priori* layers, and each layer answers a different question: L12 asks whether the generated set matches the real set as a group (fidelity), L9 asks whether the model has copied its training images (memorization), and L4 asks whether it reproduces the full variety of real images (coverage, reported as a diagnostic only). **The three axes are never averaged.** The headline M3-Score is the L12 fidelity MMD² alone — an unbiased multi-bandwidth RBF U-statistic reported with a permutation p -value, a bootstrap 95% CI, and a null-standardized separation Z — with the memorization rate and coverage reported beside it, since a single mean would conceal which of the three axes failed. The dashed path is the optional post-hoc CKA check: it never contributes to the reported score, and feeds back only to revise the layer choice *before* a subsequent evaluation. Inset plots summarize the validated behavior of the reported number (Sections IV-G and IV-A).

for new imaging modalities. The superseded adaptive design (CKA pruning and entropy-stability-uniqueness weighting) is deferred to Appendix A.

A. Notation and Pipeline Overview

Let $\mathcal{X}_r = \{x_i\}_{i=1}^N$ denote a reference set of N real radiology images and $\mathcal{X}_g = \{\hat{x}_j\}_{j=1}^N$ a set of N images produced by a generative model under evaluation. Let Φ denote the frozen RadioDINO-s16 encoder, which maps an input image to a sequence of intermediate representations across the $L = 12$ transformer blocks. The canonical M3-Score reports three axes computed in the following stages, detailed in Section III-B and summarized in Algorithm 1. (A superseded adaptive path, retained only for the post-hoc validation of Section IV-H, is *not* part of the canonical metric.)

1. **A-priori layer read-out.** Features are extracted from three fixed layers, $\ell_{\text{fid}} = 12$, $\ell_{\text{mem}} = 9$, and $\ell_{\text{cov}} = 4$, chosen by representational depth before any test outcome is observed. No layer is selected adaptively.
2. **Feature extraction.** For each axis layer ℓ , CLS-token features yield embedding matrices $E_r^{(\ell)}, E_g^{(\ell)} \in \mathbb{R}^{N \times D}$ for the real and generated sets ($D = 384$).

3. **Fidelity axis.** An unbiased multi-bandwidth RBF MMD² U-statistic quantifies the distributional divergence between $E_r^{(12)}$ and $E_g^{(12)}$, reported with a permutation p -value, null-standardized separation Z , and bootstrap 95% CI.

4. **Memorization and coverage axes.** Nearest-neighbour distances at L9 give the memorization rate; k -NN manifold precision/recall at L4 give coverage. The legacy CKA-pruning step (Algorithm 5) is run only when the optional post-hoc validation of Section IV-H is desired.

B. Canonical Three-Axis Formulation (A-Priori Layers)

The canonical M3-Score reports three axes, each read from a fixed layer of the RadioDINO-s16 encoder ($L = 12$ blocks). The layer assignments are chosen by representational position *before any evaluation outcome is observed*, and are frozen configuration constants:

$$\ell_{\text{fid}} = 12, \quad \ell_{\text{mem}} = 9, \quad \ell_{\text{cov}} = 4. \quad (1)$$

The justification is structural, not empirical: $\ell_{\text{fid}} = 12$ is the deepest semantic block (the standard choice for distributional distance, mirroring FID's use of the InceptionV3 penultimate activation); $\ell_{\text{mem}} = 9$ sits at 75% depth, where features retain

Algorithm 1 M3-Score (canonical three-axis, a-priori layers)

Require: Real set $\mathcal{X}_r$, generated set $\mathcal{X}_g$, frozen encoder Φ , a-priori layers $\ell_{\text{fid}}=12$, $\ell_{\text{mem}}=9$, $\ell_{\text{cov}}=4$

Ensure: Fidelity ($\widehat{\text{MMD}}^2$, p , [CI], Z), memorization rate, coverage (prec, rec)

- 1: **for** $\ell \in \{12, 9, 4\}$ **do**
- 2: $E_r^{(\ell)} \leftarrow \text{CLS}(\Phi, \mathcal{X}_r, \ell)$; $E_g^{(\ell)} \leftarrow \text{CLS}(\Phi, \mathcal{X}_g, \ell)$ ▷ CLS tokens; §III-C
- 3: **end for**
- 4: // Fidelity axis (L12)
- 5: $\widehat{\text{MMD}}^2 \leftarrow \text{MMD2U}(E_r^{(12)}, E_g^{(12)})$ ▷ unbiased multi-bandwidth RBF; Alg. 3
- 6: $(p, Z) \leftarrow \text{PERMTTEST}(E_r^{(12)}, E_g^{(12)}; B=500)$
- 7: [CI] $\leftarrow \text{BOOTSTRAP}(E_r^{(12)}, E_g^{(12)}; B=500)$
- 8: // Memorization axis (L9)
- 9: $\theta \leftarrow \text{median within-real 1-NN distance}$
- 10: $\text{mem} \leftarrow \frac{1}{N} \sum_j \mathbb{I}[\min_i \|E_{g,j}^{(9)} - E_{r,i}^{(9)}\|_2 < \theta]$
- 11: // Coverage axis (L4, diagnostic)
- 12: (prec, rec) $\leftarrow \text{KNN-PR}(E_r^{(4)}, E_g^{(4)}; k=5)$
- 13: **return** $\{\widehat{\text{MMD}}^2, p, [\text{CI}], Z\}$, mem, {prec, rec}

instance-level variation useful for near-duplicate detection that the final block collapses; $\ell_{\text{cov}} = 4$ sits at 33% depth, where spatial and structural diversity is encoded. This obeys a hard rule we adopt throughout: *no layer, threshold, or kernel bandwidth is selected by which choice makes a reported result look best on the test set*. We note the asymmetry in how strongly each depth is corroborated: only the fidelity layer L12 is independently supported by the CKA analysis of Section IV-H, which confirms it as the single most discriminative non-redundant layer. The memorization (L9) and coverage (L4) depths are *structural priors* chosen by representational position rather than optimised. Section IV-D reports a per-task depth ablation over all twelve blocks for both axes; it finds that L9 is neither better nor worse than any other depth under the canonical near-duplicate protocol, that no depth rescues the coverage axis, and that neither prior can be promoted to an empirically optimal choice on the evidence we have. The coverage axis is accordingly reported only as a diagnostic (Section IV-C).

a) *Fidelity axis:* Let $E_r^{(12)}, E_g^{(12)} \in \mathbb{R}^{N \times D}$ be the CLS-token features at L12. The fidelity score is the unbiased multi-bandwidth RBF MMD^2 U-statistic between them (Section III-D), with bandwidths set by the median heuristic on the pooled sample. We report it with: a permutation-test p -value (pool $E_r \cup E_g$, $B = 500$ label shuffles, one-sided $H_0 : \text{MMD}^2 = 0$); a null-standardized separation $Z = (\widehat{\text{MMD}}^2 - \mu_0)/\sigma_0$ against the permutation null (μ_0, σ_0) ; and a bootstrap 95% CI (500 resamples of r and g independently with replacement). We caution that Z is reported only as a separation magnitude, *not* as a Gaussian z -score: under H_0 the unbiased MMD^2 null variance σ_0^2 shrinks as $O(1/N)$ while the observed statistic converges to a positive constant, so Z grows approximately linearly with N . The permutation null is moreover strongly non-Gaussian, so Z does not map to a

normal tail probability; inference rests on the permutation p -value and the bootstrap CI.

b) *Memorization axis:* Using L9 features, for each generated image we compute its 1-NN L2 distance to the real set, $d_j = \min_i \|E_{g,j}^{(9)} - E_{r,i}^{(9)}\|_2$. Let θ be the median within-real 1-NN distance (excluding self). The memorization rate is $\frac{1}{N} \sum_j \mathbb{I}[d_j < \theta]$: the fraction of generated images closer to a real image than a typical real-real neighbour gap, flagging candidate near-duplicates.

c) *Coverage axis:* Using L4 features, we compute k-NN manifold precision and recall ($k = 5$) [16]: precision is the fraction of generated points inside the real k-NN manifold, recall the fraction of real points inside the generated k-NN manifold. We report this axis with the explicit caveat established in Section IV-C that it is unstable under k-NN radius inflation at small N .

The adaptive CKA-pruning and entropy-stability-uniqueness machinery, deferred to Appendix A, is retained only as an *optional, post-hoc validation* tool (it independently confirms L12 is the most discriminative fidelity layer, Section IV-H); it is not part of the canonical metric and never selects the reported axes.

An earlier *adaptive* design (CKA-based layer pruning followed by an entropy-stability-uniqueness weighting of the layer distances) was abandoned in favor of the fixed a-priori layers of Section III-B, so that no layer, threshold, or weight is ever selected on test outcomes. The CKA layer-selection step it introduced is retained purely as a post-hoc check and is specified in Appendix A (Algorithm 5); that check confirms L12 as the single most discriminative fidelity layer. The abandoned aggregation and weighting rules are not reproduced here, as no reported result uses them.

C. Feature Extraction via Frozen RadioDINO

a) *Encoder architecture:* RadioDINO [25] is a family of Vision Transformer encoders [33] pre-trained on 838,000 radiology images spanning chest radiography, computed tomography, and magnetic resonance imaging from three countries, using the DINOv2 [26] self-supervised distillation objective. We use the compact **RadioDINO-s16** variant (ViT-S/16), selected for its leading OOD discriminability among six candidate backbones (Section IV-B). The encoder has $L = 12$ transformer blocks, each with hidden dimension $D = 384$ and $H = 6$ attention heads. An input image of resolution 224×224 is partitioned into 196 non-overlapping patches of size 16×16 ; a learnable CLS token is prepended, giving a sequence of length $N_{\text{seq}} = 197$. The encoder weights are fixed throughout all M3-Score computations.

b) *Per-image feature: CLS token (canonical):* At transformer block ℓ , the encoder produces a hidden-state tensor $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B \times N_{\text{seq}} \times D}$ (and, for backbones that expose it, an attention tensor $\mathbf{A}^{(\ell)} \in \mathbb{R}^{B \times H \times N_{\text{seq}} \times N_{\text{seq}}}$), where B is the mini-batch size and $b \in \{1, \dots, B\}$ indexes an image within the batch. *For the canonical RadioDINO-s16 backbone (loaded via `timm` [34]), the per-image feature at block ℓ is the CLS token $f_b^{(\ell)} = \mathbf{H}_{b,0,:}^{(\ell)} \in \mathbb{R}^D$, consistent with Section III-B and with the standard practice of reading a ViT's aggregated*

representation from its class token. This is the configuration used for *all reported M3-Score results*, and it is what Fig. 1 depicts. Stacking $f_b^{(\ell)}$ over the N images of $\mathcal{X}$ yields the embedding matrix $E^{(\ell)} \in \mathbb{R}^{N \times D}$.

c) *Attention-aware pooling (alternative for transformers backbones)*: For HuggingFace transformers [35] ViT backbones that additionally expose per-head attention (e.g. Microsoft rad-dino, DINOv2), M3 supports an *attention-aware* pooling as an alternative to the CLS token; we describe it here for completeness and evaluate it in the backbone comparison, but it is *not* used for the RadioDINO-s16 numbers reported in this paper. Under this alternative, a weighted pooling is applied using the attention distribution that the CLS token assigns to each patch token. Let $\bar{\mathbf{A}}^{(\ell)} \in \mathbb{R}^{B \times N_{\text{seq}} \times N_{\text{seq}}}$ be the head-averaged attention matrix,

$$\bar{\mathbf{A}}^{(\ell)} = \frac{1}{H} \sum_{h=1}^H \mathbf{A}_{b,h,:}^{(\ell)}. \quad (2)$$

The CLS-to-patch attention weights are defined as

$$\alpha_{b,p}^{(\ell)} = \bar{\mathbf{A}}_{b,0,p+1}^{(\ell)}, \quad p = 1, \dots, N_{\text{seq}} - 1. \quad (3)$$

The index $p + 1$ selects patch positions only; the CLS self-attention weight $\bar{\mathbf{A}}_{b,0,0}^{(\ell)}$ is excluded, since DINOv2-style encoders produce systematically elevated self-attention at the CLS position that would otherwise dominate the pooled vector [36]. The weights are ℓ_1 -normalized per image:

$$\tilde{\alpha}_{b,p}^{(\ell)} = \frac{\alpha_{b,p}^{(\ell)}}{\sum_{p'=1}^{N_{\text{seq}}-1} \alpha_{b,p'}^{(\ell)} + \varepsilon}, \quad \varepsilon = 10^{-8}. \quad (4)$$

The attention-pooled per-image feature vector at layer ℓ for image b is

$$f_b^{(\ell), \text{attn}} = \sum_{p=1}^{N_{\text{seq}}-1} \tilde{\alpha}_{b,p}^{(\ell)} \cdot \mathbf{H}_{b,p+1,:}^{(\ell)} \in \mathbb{R}^D, \quad (5)$$

which replaces the canonical CLS feature $f_b^{(\ell)}$ only when this alternative backbone path is selected. Algorithm 2 states the complete attention-aware procedure, including mini-batch processing and accumulation across batches.

D. Per-Layer Unbiased MMD²

a) *Feature normalization*: RadioDINO intermediate hidden states returned by the transformer blocks have ℓ_2 norms in the range $[10^2, 3 \times 10^2]$ depending on layer depth. Without correction, this arbitrary scale dominates the pairwise Euclidean distances that enter the Gaussian kernel and destabilizes the median-heuristic bandwidth, rendering MMD² estimates incomparable across layers and datasets. To eliminate this dependence on the arbitrary scale of intermediate hidden states, each embedding vector is projected onto the unit sphere prior to kernel evaluation:

$$\hat{e} = \frac{e}{\|e\|_2 + \varepsilon}, \quad \varepsilon = 10^{-8}. \quad (6)$$

Algorithm 2 Attention-aware feature extraction

Require: Encoder Φ , image set $\mathcal{X}$, target layer set $\mathcal{L}$, mini-batch size $B = 16$
Ensure: $\{E^{(\ell)}\}_{\ell \in \mathcal{L}}$, $E^{(\ell)} \in \mathbb{R}^{|\mathcal{X}| \times D}$

- 1: **buf**[ℓ] $\leftarrow \emptyset$ for all $\ell \in \mathcal{L}$
- 2: **for** each mini-batch $\mathcal{B} \subseteq \mathcal{X}$ of size $\min(B, |\mathcal{X}|)$ **do**
- 3: $\mathcal{B} \leftarrow$ apply RadioDINO image processor to $\mathcal{B}$
- 4: $\{\mathbf{H}^{(\ell)}\}, \{\mathbf{A}^{(\ell)}\} \leftarrow \Phi(\mathcal{B})$ $\triangleright$ all layers, hidden states and attentions
- 5: **for** each $\ell \in \mathcal{L}$ **do**
- 6: $\bar{\mathbf{A}}^{(\ell)} \leftarrow \frac{1}{H} \sum_{h=1}^H \mathbf{A}_{:,h,:}^{(\ell)}$ $\triangleright (B \times N_{\text{seq}} \times N_{\text{seq}})$
- 7: $\alpha^{(\ell)} \leftarrow \bar{\mathbf{A}}_{:,0,1:}^{(\ell)}$ $\triangleright$ CLS-to-patch weights, $(B \times (N_{\text{seq}} - 1))$
- 8: $\tilde{\alpha}^{(\ell)} \leftarrow \alpha^{(\ell)} / (\sum_p \alpha_{:,p}^{(\ell)} + \varepsilon)$ $\triangleright \ell_1$ -normalize, Eq. (4)
- 9: $\mathbf{P}^{(\ell)} \leftarrow \mathbf{H}_{:,1,:}^{(\ell)}$ $\triangleright$ patch tokens, $(B \times (N_{\text{seq}} - 1) \times D)$
- 10: $f^{(\ell)} \leftarrow \sum_p \tilde{\alpha}_{:,p}^{(\ell)} \cdot \mathbf{P}_{:,p,:}^{(\ell)}$ $\triangleright$ weighted sum, Eq. (5), $(B \times D)$
- 11: **buf**[ℓ] $\leftarrow$ **buf**[ℓ] $\cup \{f^{(\ell)}\}$
- 12: **end for**
- 13: **end for**
- 14: **for** each $\ell \in \mathcal{L}$ **do**
- 15: $E^{(\ell)} \leftarrow$ concatenate(**buf**[ℓ], axis=0)
- 16: **end for**
- 17: **return** $\{E^{(\ell)}\}_{\ell \in \mathcal{L}}$

After this projection, $\|\hat{e}\|_2 = 1$ and the inner product $\hat{u}^\top \hat{v}$ equals the cosine similarity of u and v , which lies in $[-1, 1]$. This normalization is applied to the rows of both $E_r^{(\ell)}$ and $E_g^{(\ell)}$ before any kernel computation. The CKA layer selection step (Appendix A) operates on the raw (unnormalized) features, since CKA is scale-invariant and measures representational geometry rather than distributional distance.

b) *Kernel definition*: The distributional divergence between the unit-normalized real and generated embedding sets is measured by the Maximum Mean Discrepancy [27] under a *multi-bandwidth Gaussian RBF* kernel. For two unit-norm vectors $u, v \in \mathbb{R}^D$,

$$k(u, v) = \frac{1}{|\mathcal{B}|} \sum_{\sigma^2 \in \mathcal{B}} \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right), \quad (7)$$

the average of five Gaussian kernels whose squared bandwidths

$$\mathcal{B} = \left\{\frac{1}{4}, \frac{1}{2}, 1, 2, 4\right\} \cdot \sigma_{\text{med}}^2 \quad (8)$$

form a geometric ladder around a base scale σ_{med}^2 set by the *median heuristic* [27]: σ_{med}^2 is the median of the pairwise squared Euclidean distances over the pooled sample $\hat{R} \cup \hat{G}$ (upper triangle, self-distances excluded). Averaging over a ladder of bandwidths makes the kernel sensitive to distributional differences at several scales simultaneously and removes the need to tune a single bandwidth, while the median anchor adapts the kernel to the geometry of each feature cloud. On the unit sphere, $\|u - v\|_2^2 = 2(1 - u^\top v) \in [0, 4]$, so every kernel

Algorithm 3 Unbiased multi-bandwidth RBF MMD²

Require: Real and generated embedding matrices at a single layer ℓ , $E_r, E_g \in \mathbb{R}^{N \times D}$ (Section III-A); bandwidth scales $S = \{0.25, 0.5, 1, 2, 4\}$

Ensure: $\widehat{\text{MMD}}^2 \in [0, \infty)$

```

1:  $\hat{E}_{r,i,:} \leftarrow E_{r,i,:}/(\|E_{r,i,:}\|_2 + \varepsilon)$ ;  $\hat{E}_{g,j,:} \leftarrow E_{g,j,:}/(\|E_{g,j,:}\|_2 + \varepsilon)$   $\triangleright \ell_2$ -normalize rows, Eq. (6)
2:  $D_{ij}^{rr} \leftarrow \|\hat{E}_{r,i} - \hat{E}_{r,j}\|_2^2$ ;  $D_{ij}^{gg} \leftarrow \|\hat{E}_{g,i} - \hat{E}_{g,j}\|_2^2$ ;  $D_{ij}^{rg} \leftarrow \|\hat{E}_{r,i} - \hat{E}_{g,j}\|_2^2$ 
3:  $\sigma_{\text{med}}^2 \leftarrow \text{median}(\{D_{ij}^{\cup} : i < j\})$   $\triangleright$  median heuristic on  $\hat{E}_r \cup \hat{E}_g$ , Eq. (8)
4:  $A \leftarrow 0$ 
5: for each scale  $s \in S$  do
6:    $\gamma \leftarrow 1/(2s\sigma_{\text{med}}^2)$ 
7:    $K^{rr} \leftarrow \exp(-\gamma D^{rr})$ ;  $K^{gg} \leftarrow \exp(-\gamma D^{gg})$ ;  $K^{rg} \leftarrow \exp(-\gamma D^{rg})$ 
8:    $T_{rr} \leftarrow (\sum_{i,j} K_{ij}^{rr} - \sum_i K_{ii}^{rr})/(N(N-1))$   $\triangleright$  diagonal excluded
9:    $T_{gg} \leftarrow (\sum_{i,j} K_{ij}^{gg} - \sum_i K_{ii}^{gg})/(N(N-1))$ 
10:   $T_{rg} \leftarrow (\sum_{i,j} K_{ij}^{rg})/N^2$ 
11:   $A \leftarrow A + (T_{rr} + T_{gg} - 2T_{rg})$ 
12: end for
13:  $\widehat{\text{MMD}}^2 \leftarrow \max(A/|S|, 0)$   $\triangleright$  Eq. (9); clamp removes finite- $N$  negative bias
14: return  $\widehat{\text{MMD}}^2$ 
```

value lies in $(0, 1]$ and the resulting MMD^2 is bounded and directly comparable across datasets, analogous to KID [12].

c) *Unbiased estimator:* Let $\hat{R} \in \mathbb{R}^{N \times D}$ and $\hat{G} \in \mathbb{R}^{N \times D}$ denote the ℓ_2 -normalized real and generated embedding matrices at the fidelity layer. Define the kernel matrices $K_{ij}^{rr} = k(\hat{R}_{i,:}, \hat{R}_{j,:})$, $K_{ij}^{gg} = k(\hat{G}_{i,:}, \hat{G}_{j,:})$, $K_{ij}^{rg} = k(\hat{R}_{i,:}, \hat{G}_{j,:})$ using the multi-bandwidth RBF kernel of Eq. (7) (by linearity, averaging the per-bandwidth U-statistics is equivalent to forming the U-statistic of the averaged kernel). The unbiased MMD^2 estimator is

$$\widehat{\text{MMD}}_\ell^2 = \frac{1}{N(N-1)} \sum_{i \neq j} K_{ij}^{rr} + \frac{1}{N(N-1)} \sum_{i \neq j} K_{ij}^{gg} - \frac{2}{N^2} \sum_{i,j} K_{ij}^{rg}, \quad (9)$$

clamped below at zero to remove finite-sample negative artefacts; this introduces a small one-sided bias only in the regime where the true MMD^2 is near zero, and the estimator remains asymptotically unbiased. Unlike FID, Eq. (9) makes no parametric assumption on the distribution of embeddings, is an (asymptotically) unbiased estimator of the true population MMD^2 , and its expectation is zero if and only if the real and generated distributions are identical under k . Algorithm 3 states the computation explicitly; we refer to it as MMD^2_{U} where it is invoked from other algorithms.

E. Per-Image Score

The population-level score S summarizes distributional divergence across all N generated images. A complementary per-image score enables ranking of individual generated images by their distance from the real distribution, which is

Algorithm 4 Per-image quality ranking

Require: Generated L12 embeddings $E_g^{(12)}$, real L12 embeddings $E_r^{(12)}$

Ensure: Index permutation π such that $s_{\pi(1)} \geq s_{\pi(2)} \geq \dots \geq s_{\pi(N)}$

```

1:  $\ell_2$ -normalize all rows of  $E_r^{(12)}$  and  $E_g^{(12)}$   $\triangleright$  Eq. (6)
2:  $\hat{\mu}_r^{(12)} \leftarrow \frac{1}{N} \sum_{i=1}^N \hat{E}_{r,i}^{(12)}$   $\triangleright$  real centroid at L12
3: for  $j = 1, \dots, N$  do
4:    $s_j \leftarrow \|\hat{E}_{g,j}^{(12)} - \hat{\mu}_r^{(12)}\|_2$   $\triangleright$  Eq. (10)
5: end for
6:  $\pi \leftarrow \text{argsort}(s_1, \dots, s_N, \text{descending})$ 
7: return  $\pi$ 
```

absent from FID, KID, and FRD. Consistent with the canonical fidelity axis, the per-image score is defined on the final (L12) layer alone, so it requires no adaptive weighting. For an image $\hat{x}_j \in \mathcal{X}_g$ with ℓ_2 -normalized feature vector $\hat{E}_{g,j}^{(12)}$, the score is the distance to the real centroid

$$s(\hat{x}_j) = \|\hat{E}_{g,j}^{(12)} - \hat{\mu}_r^{(12)}\|_2, \quad \hat{\mu}_r^{(12)} = \frac{1}{N} \sum_{i=1}^N \hat{E}_{r,i}^{(12)}, \quad (10)$$

where $\hat{\mu}_r^{(12)}$ is the empirical centroid of the ℓ_2 -normalized real L12 embeddings. This is exactly the per-image quantity used in the anomaly-correlation experiment (Section IV-B, Fig. 6(b)). Images with large $s(\hat{x}_j)$ lie further from the real distribution at the semantic (L12) scale. (An unweighted average over the three canonical layers $\{12, 9, 4\}$ is an equally valid variant; we default to L12 for direct correspondence with the fidelity axis.) The centroid $\hat{\mu}_r^{(\ell)}$ is a by-product of the feature extraction step, so no additional computation is required beyond what is performed for S . Algorithm 4 states the ranking procedure.

F. Computational Complexity

a) *Time complexity:* The dominant cost is feature extraction, which requires a forward pass through Φ for all N images. Self-attention at each block costs $O(N_{\text{seq}}^2 \cdot D)$ per image, giving $O(N \cdot N_{\text{seq}}^2 \cdot D)$ total for a single set. The CKA layer selection step processes N_r images over all L layers and requires $O(L^2 \cdot N_r^2)$ gram-matrix inner products; with $N_r = 20$ and $L = 12$, this is negligible relative to the main extraction pass. The ℓ_2 -normalization step costs $O(N \cdot M \cdot D)$ and the MMD^2 computation at each retained layer costs $O(N^2)$ kernel evaluations; both are small relative to feature extraction at $N = 500$. Total wall-clock time at $N = 500$ on a single NVIDIA A6000 GPU is approximately 7.87s with the three a-priori read-out layers, compared to 80.92s for Manifold Precision/Recall at the same sample size. Feature extraction requires a single forward pass through Φ regardless of M ; the additional layers contribute only hook-collection overhead, so wall-clock time scales sub-linearly with M .

b) *Memory:* Embedding matrices for both sets at the M read-out layers require $2MND \times 4$ bytes in single-precision floating point. At the canonical three-axis configuration ($M = 3$ a-priori layers, $N = 500$, $D = 384$), this amounts to

approximately 4.4 MB ($2 \times 3 \times 500 \times 384 \times 4$ bytes), well within the capacity of any contemporary evaluation GPU. Even the post-hoc CKA pass at its least aggressive setting ($\tau = 0.99$, $M = 11$ layers) requires only about 16 MB.

c) Hyperparameters: Because the canonical metric fixes its read-out layers a priori (Section III-B), it has *no* layer-selection or weighting hyperparameter. The fidelity axis uses a fixed, median-anchored five-bandwidth RBF ladder (Eq. (8)) and a permutation count B (default 500); the only deployment guidance is the sample size N (≥ 100 for monitoring, 500 for publication). The optional post-hoc CKA validation retains two legacy parameters, the redundancy threshold τ (default 0.80) and sub-batch count K (default 5), whose sensitivity is reported in Section IV-H.

G. Application to New Imaging Modalities

The M3-Score requires no retraining and no modality-specific configuration when applied to a *radiology* modality outside the BraTS brain MRI validation domain (Section I-D). The procedure is as follows.

1. **Evaluation set preparation.** Assemble $N \geq 100$ real and generated images from the target modality. Images are resized to 224×224 and normalized to $[0, 1]$ by the RadioDINO preprocessing pipeline.
2. **A-priori layer read-out.** Extract CLS-token features at the three fixed layers $\ell_{\text{fid}} = 12$, $\ell_{\text{mem}} = 9$, $\ell_{\text{cov}} = 4$ (Section III-B). No layer is selected adaptively, so the configuration is identical across modalities and reproducible without a calibration set.
3. **Axis computation.** Compute the fidelity MMD^2 at L12 (Algorithm 3) together with its permutation p -value, bootstrap 95% CI, and null-standardized separation Z ; the L9 memorization rate; and the L4 coverage precision/recall.
4. **Axis interpretation.** A large fidelity MMD^2 with significant Z indicates a distributional gap; a non-zero memorization rate flags near-duplicate leakage; low coverage indicates limited diversity (subject to the small- N caveat of Section IV-C).
5. **Per-image ranking.** Optionally execute Algorithm 4 to rank generated images by $s(\hat{x}_j)$ for downstream review.

For training-time monitoring, $N \geq 100$ images are sufficient to obtain stable estimates; for publication-grade evaluation, $N \geq 500$ is recommended. Increasing N reduces the variance of the fidelity $\widehat{\text{MMD}}^2$ at a cost proportional to the additional forward passes through Φ . The optional post-hoc CKA validation (Algorithm 5) may additionally be run on $N_r = 20$ real images to confirm that L12 remains the most discriminative fidelity layer for the new modality.

IV. RESULTS AND EVALUATION

We evaluate the M3-Score under one canonical configuration applied across every experiment: RadioDINO-s16 backbone, $N = 500$ where sample size permits, multi-bandwidth RBF kernel, and the unbiased U-statistic MMD^2 . The real reference set is $N = 500$ axial BraTS slices and the generated

set is $N = 500$ images from an unconditional DDPM trained on the same distribution; representative samples of both sets are shown in Fig. 2. The fidelity, memorization, and coverage axes read from the a-priori layers L12, L9, and L4 respectively (Section III-B); no layer is selected by test outcome. We organize results into seven subsections: a noise/blur sanity check, OOD and cross-modality discrimination (including the CMMD failure), three-axis validation, a generation-quality ladder, pathology-region masking, sample efficiency, a post-hoc CKA ablation, and the permutation test. Results that do not favor M3 are reported alongside those that do. The Isolation Forest anomaly detector used in Section IV-B is trained exclusively on real image embeddings and is completely independent of the M3-Score computation.

a) Reported quantities: The tables below use the following quantities throughout. ρ is the Spearman rank correlation between a perturbation level and a metric's value; a metric that increases monotonically with degradation attains $\rho = 1$. τ is Kendall's rank correlation, used where the ordering rather than the spacing of a small number of tiers is at issue. CV is the coefficient of variation, the standard deviation across repeated subsamples divided by their mean, expressed as a percentage; lower means a metric returns a more repeatable value at a given N . Z is the null-standardized separation defined in Section III-B, reported as a magnitude of separation from the permutation null rather than as a Gaussian z -score. α -Precision is the fraction of generated samples that fall inside the k -NN manifold of the real set ($k = 5$), and β -Recall the fraction of real samples that fall inside the k -NN manifold of the generated set; the two measure fidelity and coverage respectively, and both lie in $[0, 1]$ with higher being better [16], [23]. Δ denotes a paired difference between two rows of the same table, reported with its bootstrap 95% CI.

A. Baseline Sanity Check: Noise and Blur Robustness

A necessary condition for any evaluation metric is monotonic response to known image degradation. We inject progressive Gaussian noise ($\sigma \in \{0.0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$) and Gaussian blur (radius $r \in \{0, 1, 2, 3, 4\}$ pixels) into the real image set and compute all nine metrics between the clean reference and the perturbed set. FRD is excluded due to incompatibility between the pyradiomics Laplacian-of-Gaussian filter and BraTS axial image dimensions [32]. Table II reports Spearman rank correlations between perturbation level and metric value, together with 95% confidence intervals via Fisher z -transformation.

a) Noise degradation: At $\sigma = 0$ (clean versus clean), M3 returns exactly 0.0, consistent with the unbiased MMD^2 estimator whose expectation is zero under identical distributions. All nine metrics achieve perfect Spearman correlation ($|\rho| = 1.0$, $p < 0.001$) under Gaussian noise, confirming that M3 responds correctly to progressive pixel-level degradation (Fig. 3 and Fig. 4). As noise increases to $\sigma = 0.5$, the fidelity MMD^2 rises to 0.607, FID to 521.96, and KID to 0.868, while SSIM collapses to 0.031 and PSNR to 9.29 dB (raw unnormalized values; Fig. 3 and Fig. 4 display all metrics normalized to $[0, 1]$ for visual comparison).

Representative BraTS axial slices: real vs. DDPM-generated

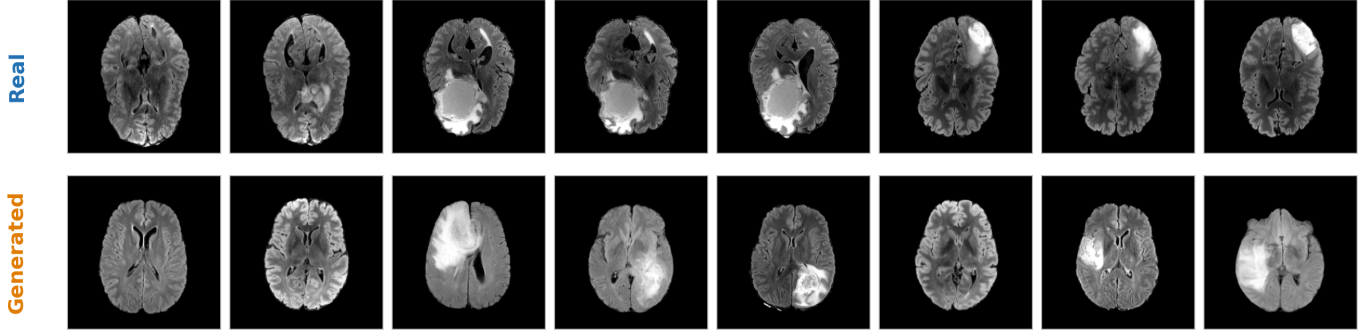


Fig. 2: Representative axial BraTS slices: real (top) and unconditional DDPM-generated (bottom). Both sets exhibit plausible tissue boundaries, ventricles, and FLAIR-hyperintense lesions, and are difficult to separate by visual inspection, which motivates a rigorous distributional metric with native significance testing rather than a qualitative check. Images are shown at the 224×224 resolution fed to the encoder.

TABLE II: Spearman rank correlation ρ between perturbation level and metric value ($N = 7$ noise levels; $N = 5$ blur radii). $\uparrow$: metric should increase with perturbation. $\downarrow$: metric should decrease. All $p < 0.001$ except α -Precision and β -Recall under blur ($p = 0.005$).

Metric	Dir.	Spearman ρ	
		Noise	Blur
M3-Score (ours)	$\uparrow$	1.000	1.000
FID (Inception)	$\uparrow$	1.000	1.000
KID (Inception)	$\uparrow$	1.000	1.000
SSIM	$\downarrow$	-1.000	-1.000
PSNR	$\downarrow$	-1.000	-1.000
MS-SSIM	$\downarrow$	-1.000	-1.000
LPIPS	$\uparrow$	1.000	1.000
α -Precision	$\downarrow$	-0.982	-0.975
β -Recall	$\downarrow$	-0.982	-0.975

b) Blur degradation: Under Gaussian blur the M3-Score achieves $\rho = 1.000$ ($p < 0.001$), identical to FID and KID, and the sequence is strictly monotone across all five radii. The α -Precision metric reveals that blur is substantially more destructive to manifold membership than noise at comparable image quality. At blur radius $r = 2$ (PSNR = 27.88 dB), α -Precision collapses to 0.006: virtually no perturbed images remain within the real clinical manifold. At Gaussian noise $\sigma = 0.05$ (PSNR = 27.96 dB, similar quality), α -Precision remains 0.384. This confirms that the real BraTS image manifold is defined primarily by high-frequency texture structure that blur destroys while additive noise partially preserves.

B. Unsupervised Anomaly Detection and OOD Correlation

A central claim of the M3-Score is that its fidelity axis identifies clinically anomalous generated images without requiring explicit anomaly labels. We test this by fitting an Isolation Forest [37] on embeddings of the 500 real images and scoring each of the 500 generated images independently of M3. The fidelity axis uses the a-priori layer $\ell_{\text{fid}} = 12$ (Section III-B); no adaptive weighting is involved.

TABLE III: Spearman correlation between per-image Isolation Forest anomaly score and per-image distance measures ($N = 500$ generated images). 95% CIs via Fisher z -transformation.

Distance measure	ρ	p	95% CI
M3-Score (RadioDINO, L2-norm)	+0.238	7.5×10^{-8}	[0.153, 0.319]
FID proxy (InceptionV3)	+0.211	2.0×10^{-6}	[0.126, 0.293]
Pixel MSE	-0.199	7.5×10^{-6}	[-0.282, -0.113]

a) OOD rate: With contamination threshold 0.05, 16.0% of generated images (80 of 500) are flagged as out-of-distribution. The mean anomaly score of generated images (-0.020) is systematically higher than that of real images (-0.047), confirming that the generated distribution occupies lower-probability regions of the real clinical manifold.

b) Per-image score correlation: Table III reports Spearman correlations between the Isolation Forest anomaly score and three per-image distance measures.

M3 per-image scores show the highest point correlation with the independent anomaly detector ($\rho = +0.238$, $p = 7.5 \times 10^{-8}$), slightly above the FID proxy ($\rho = +0.211$) despite using a completely different backbone and distance measure. However, the two 95% CIs overlap ([0.153, 0.319] vs. [0.126, 0.293]), so this per-image ranking difference is *not* statistically significant: at the per-image level M3 and the FID proxy are comparable, and M3's advantage over FID is established at the *distributional* level instead (ROC-AUC and the permutation test, Section IV-B), not in per-image anomaly ranking. Pixel MSE exhibits a significant *negative* correlation ($\rho = -0.199$), confirming the well-documented failure of pixel-space distances to capture clinical anomaly: an image can be pixel-similar to the real mean while being semantically implausible in clinical terms (Fig. 6(b)).

c) Binary distributional separation: To formally quantify whether RadioDINO or InceptionV3 features provide better separation between real and generated images, we treat the centroid-L2 distance as a binary classifier score (real = 0, generated = 1) and compute ROC-AUC and PR-AUC for both backbones. Tables IV and V report the results.

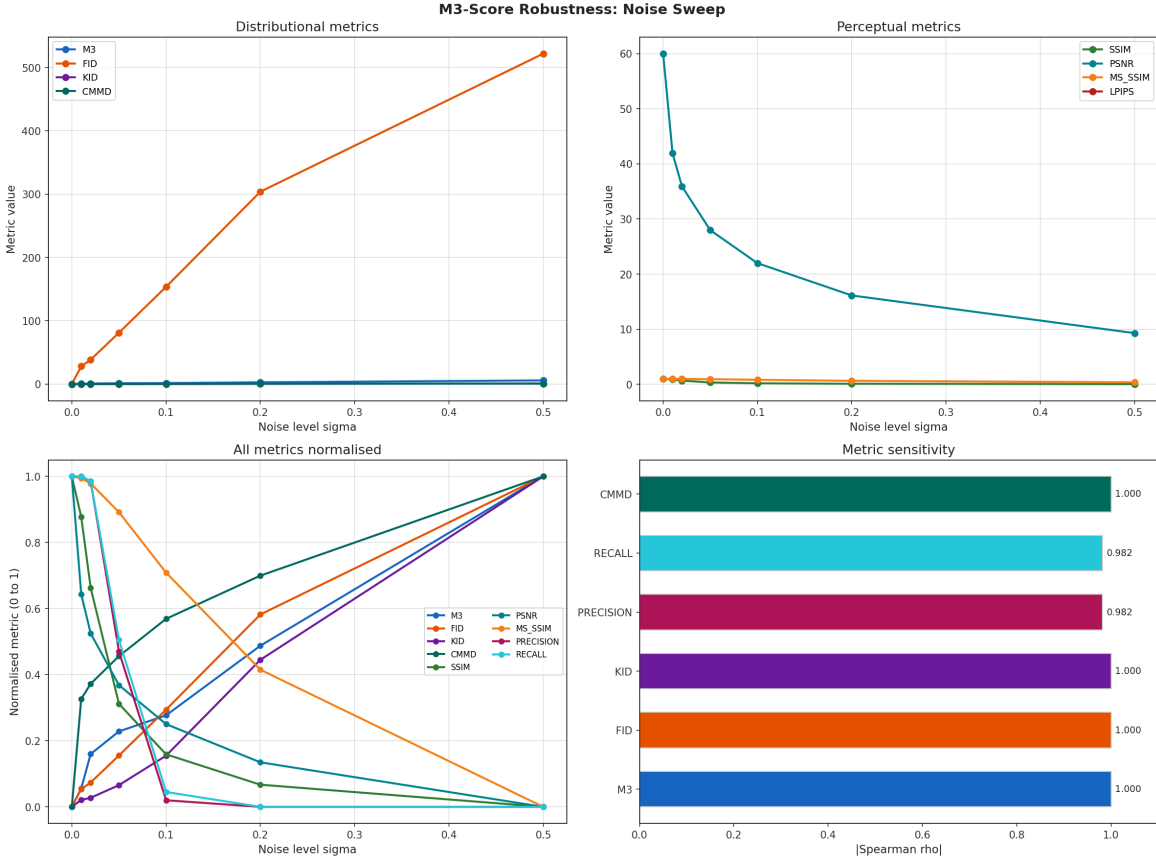


Fig. 3: **Gaussian noise sweep** ($\sigma \in [0, 0.5]$, $N = 200$ BraTS images). The composite panel shows four views: (a) distributional metrics (M3/FID/KID) in raw units, (b) perceptual metrics (SSIM/PSNR/MS-SSIM/LPIPS), (c) all metrics normalized to $[0, 1]$ for visual comparison, and (d) $|\text{Spearman } \rho|$ bar chart. All nine metrics achieve $|\rho| = 1.00$ ($p < 0.001$); the normalized panel (c) shows M3 and SSIM/PSNR diverge earliest, indicating higher sensitivity at low perturbation levels. Note: raw metric values cited in text (e.g. fidelity $\text{MMD}^2 = 0.607$, $\text{FID} = 521.96$ at $\sigma = 0.5$) correspond to unnormalized panel (a)/(b); panel (c) shows the same data rescaled.

TABLE IV: **Metric-backbone comparison** at the canonical $N = 500$: OOD ROC-AUC (BraTS real vs. DDPM-generated, centroid- L_2 score). The M3 fidelity backbone is compared against the backbones that underlie FID and CMMD.

Metric (backbone)	OOD ROC-AUC ($N=500$)
M3 (RadioDINO-s16, ours)	0.976
FID (InceptionV3)	0.809
CMMD (CLIP ViT-L/14)	0.659

The RadioDINO-s16 fidelity axis achieves ROC-AUC = 0.976 at $N = 500$ versus 0.809 for InceptionV3 (the FID backbone) and 0.659 for CLIP (the CMMD backbone), a +20.6% and +48.1% relative margin respectively (Fig. 6(c)). The $N = 100$ six-backbone selection screen (Table V) confirms the choice: RadioDINO-s16 (0.995) leads the larger Microsoft rad-dino ViT-B/14 (0.913) and generic DINOv2 (0.968). The most “distinctive” backbone by inter-layer CKA (rad-dino, $L1 \leftrightarrow L12$ CKA = 0.203) is the *worst* OOD discriminator, so layer distinctiveness does not predict discriminative power; we select by OOD-AUC, not by CKA structure.

TABLE V: **Backbone selection screen** for the M3 fidelity axis: OOD ROC-AUC across candidate medical/general encoders (BraTS real vs. DDPM-generated, $N = 100$). RadioDINO-s16 is selected for all other experiments; it also leads at $N = 500$ (Table IV), so the choice is not an artifact of the screening sample size.

Candidate backbone	OOD ROC-AUC ($N=100$)
RadioDINO-s16 (ours)	0.995
DINOv2-base (ViT-B/14)	0.968
RadImageNet-ResNet50	0.954
PubMed-CLIP	0.921
Microsoft rad-dino (ViT-B/14)	0.913

d) CMMD structural failure on medical OOD ordering:

The most consequential comparison is not the aggregate AUC but a specific ordering failure. We evaluate three tiers against BraTS real ($N = 500$, with bootstrap 95% CIs over 500 resamples): a same-domain DDPM generator, a weaker same-domain (WDM-3D) generator, and a full cross-modality shift to chest/lung CT (LIDC). CMMD here follows the standard protocol of Jayasumana *et al.* [30]: OpenAI

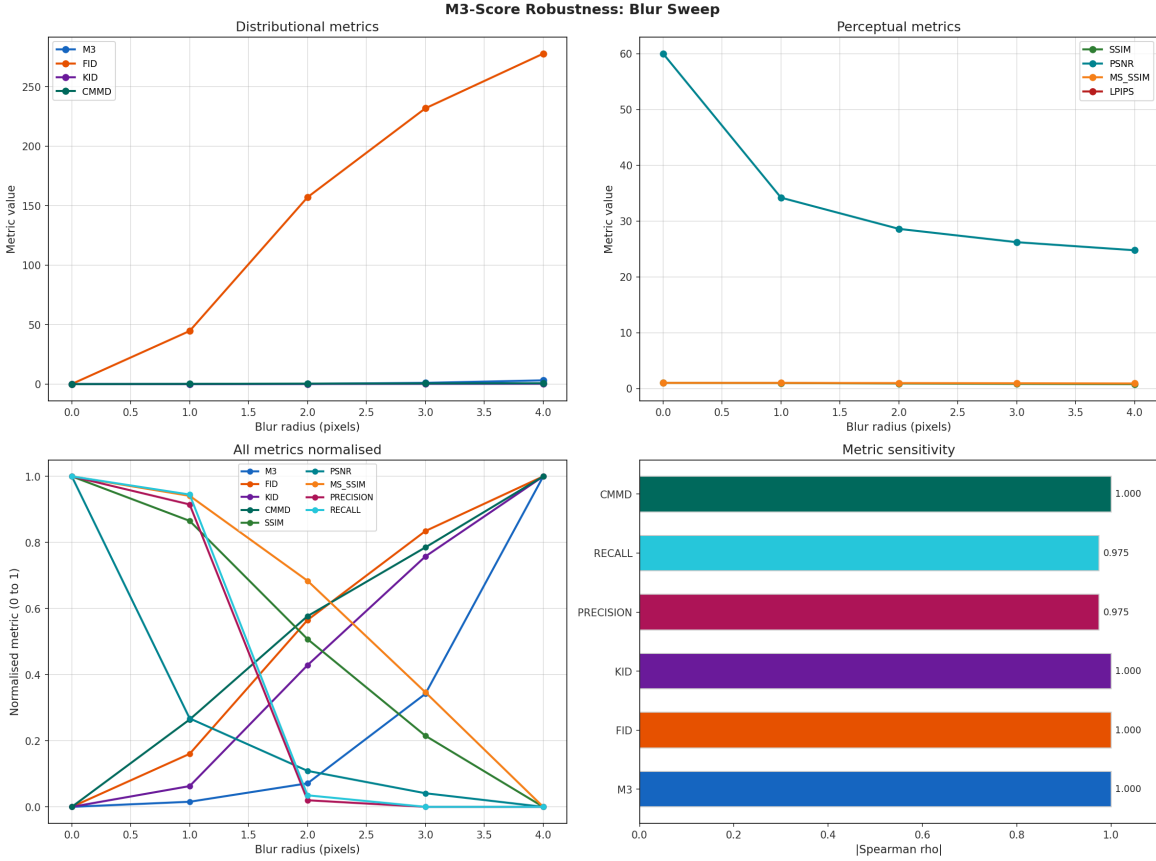


Fig. 4: **Gaussian blur sweep** ($r \in \{0, 1, 2, 3, 4\}$ pixels, $N = 200$ BraTS images). The composite panel shows four views: (a) distributional metrics (M3/FID/KID) in raw units, (b) perceptual metrics (SSIM/PSNR/MS-SSIM/LPIPS), (c) all metrics normalized to $[0, 1]$ for visual comparison, and (d) $|\text{Spearman } \rho|$ bar chart. M3-Score achieves $\rho = 1.000$ ($p < 0.001$). α -Precision collapses to 0.006 by $r = 2$, while distributional metrics remain monotonic, revealing that the real BraTS manifold is defined by high-frequency texture structure that Gaussian blur selectively destroys.

CLIP ViT-L/14 image features and an unbiased Gaussian-RBF MMD² with the median-heuristic bandwidth computed on the pooled real-plus-generated sample, so the inversion below cannot be attributed to a non-standard estimator or bandwidth. The correct order is DDPM < WDM-3D < CT, since a different modality should be the most distant; the three tiers are illustrated in Fig. 5. The M3 fidelity axis recovers this order with non-overlapping CIs ($0.153 < 0.248 < 0.457$; Table VI). CMMD, however, *inverts* the last two tiers: it assigns the cross-modality CT ($0.699 [0.686, 0.715]$) a *smaller* distance than the weak same-modality MRI generator ($0.738 [0.723, 0.758]$), ranking lung CT as closer to brain MRI than a poor brain-MRI generator. The paired difference $\Delta_{\text{CT-MRI}} = -0.039 [-0.059, -0.021]$ excludes zero, so this is a statistically significant rank inversion, not sampling noise; the corresponding M3 difference is $+0.210 [+0.198, +0.220]$. FID (InceptionV3) also orders the tiers correctly ($68.8 < 180.6 < 317.4$), so the failure is specific to *CLIP* features rather than to natural-image backbones in general: CLIP does not encode the radiology modality boundary that a radiology-trained backbone, and even InceptionV3, does.

e) Domain-ontology ordering: a scope limitation:

Across five test sets ordered by intuitive OOD severity, FID

TABLE VI: Cross-tier separation from BraTS real ($N = 500$; bootstrap 95% CIs for M3 and CMMD over 500 resamples). Higher = more distant, so the correct order is DDPM < WDM-3D < CT. M3 and FID order the tiers correctly; CMMD *inverts* the last two, ranking cross-modality CT as closer than the weak MRI generator ($\Delta_{\text{CT-MRI}} < 0$, CI excludes zero). FID values are point estimates (InceptionV3).

Comparison vs. BraTS real	M3 (fid.) [95% CI]	FID	CMMD
DDPM (same-domain)	0.153 [0.146, 0.162]	68.8	0.133
WDM-3D (weak same-domain)	0.248 [0.243, 0.257]	180.6	0.738
LIDC CT (cross-modality OOD)	0.457 [0.449, 0.472]	317.4	0.699
CT – WDM-3D Δ	+0.210 [+0.198, +0.220]	+136.8	−0.039 [−0.059, −0.021]

and KID achieve Kendall $\tau = 1.000$ while M3 achieves $\tau = 0.800$ ($p = 0.083$). The single inversion is that M3 ranks retinal fundus images as *further* from brain MRI than chest CT. This is not an error: under a radiology-domain ontology, CT and MRI are neighbours and fundus is out-of-domain, which RadioDINO correctly reflects. It does, however, mean M3 should not be read as a universal visual-distance metric, and we restrict its claims to the radiology domain accordingly.

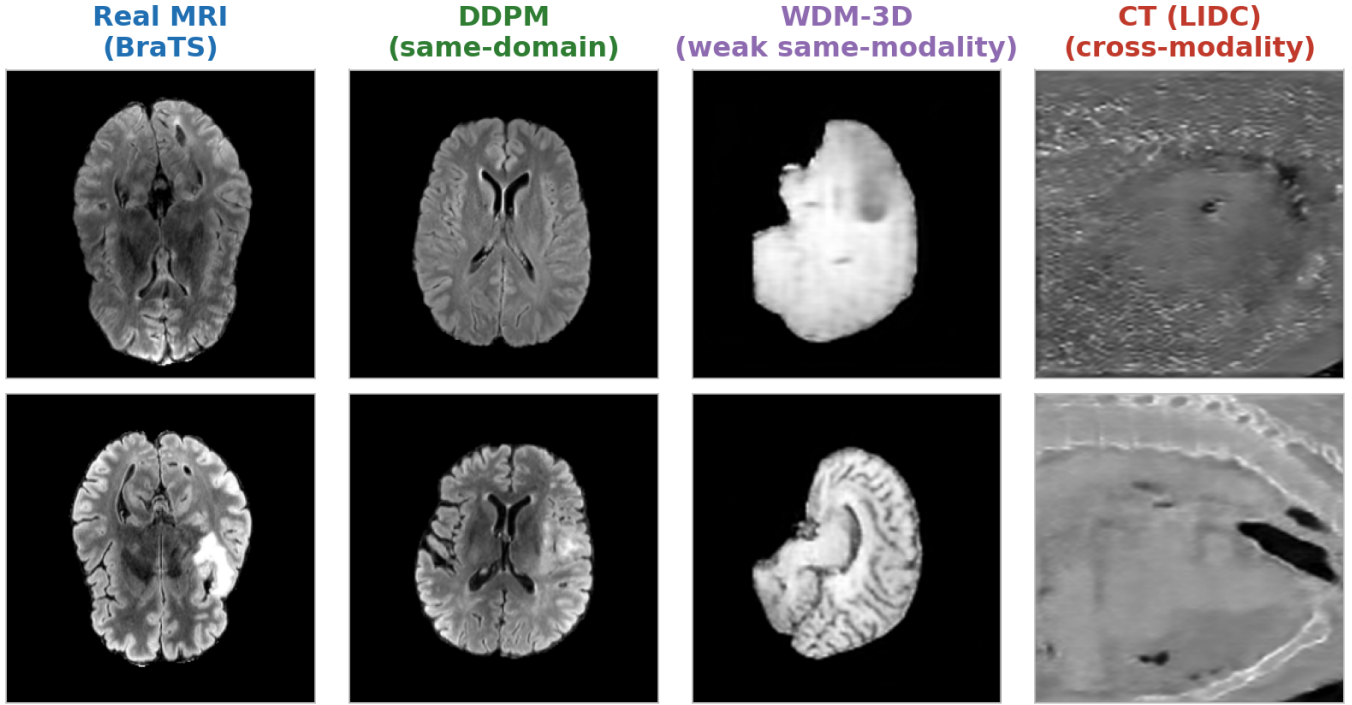


Fig. 5: The three tiers evaluated against real BraTS MRI. From left: real MRI; a same-domain DDPM generator; a weaker same-modality WDM-3D generator (visibly lower-quality but still brain MRI); and a cross-modality LIDC chest CT scan, which is a different modality and anatomy. A faithful metric should rank the CT as the most distant from real brain MRI. M3 and FID do; CMMD inverts the last two, ranking the CT (right) as closer to real MRI than the weak WDM-3D brain generator (Table VI).

C. Three-Axis Validation

We validate the memorization and coverage axes by injecting the specific failure each is designed to detect and checking that the corresponding axis, and only that axis, responds. Features are read from the a-priori layers $\ell_{\text{mem}} = 9$ and $\ell_{\text{cov}} = 4$ (Section III-B); $N = 150$ BraTS real and DDPM-generated images.

a) Memorization axis (validated): We replace a known fraction of generated images with near-duplicates of real images (Gaussian jitter, $\sigma = 5 \times 10^{-4}$) at injection rates $\{0.0, 0.1, 0.2, 0.3, 0.5\}$. The L9 memorization rate recovers the injection rate *exactly* (0.00, 0.10, 0.20, 0.30, 0.50; Table VII), confirming the axis flags near-duplicates at the intended depth. As expected, the fidelity MMD² at L12 simultaneously falls (from 0.187 to 0.043), because injecting genuine real images into the generated set also moves the generated distribution toward the real one; the two axes are therefore correlated under this perturbation rather than orthogonal.

b) Coverage axis: a negative result: Probing the coverage axis with mode-drop (truncating the generated set by $\{0, 10, 20, 30, 50\}\%$) does *not* produce a monotone decrease in L4 recall; recall instead drifts upward (0.18 $\rightarrow$ 0.28) as the generated set shrinks. The mechanism is well understood: with fewer generated points, each point’s k -NN radius inflates, the generated manifold balls expand, and they cover more real points despite reduced diversity. This is a known instability of k -NN precision/recall at small N [16], [24]. We therefore

TABLE VII: Memorization-axis injection test ($N = 150$). The L9 memorization rate recovers the injected near-duplicate fraction exactly; L12 fidelity also responds because injected real images shift the generated distribution.

Injection rate	Mem. rate (L9)	Fidelity MMD ² (L12)	Coverage recall (L4)
0.0	0.00	0.187	0.18
0.1	0.10	0.150	0.89
0.2	0.20	0.116	0.93
0.3	0.30	0.090	1.00
0.5	0.50	0.043	0.99

report the coverage axis as a diagnostic under active development, not a validated claim, and recommend it only with $N \gtrsim 500$ and semantically clustered mode-drop. Precision is identically 0.0 in our setting, consistent with the OOD result that generated points lie outside the real k -NN manifold at this checkpoint.

D. Layer-Depth Ablation for the Memorization and Coverage Axes

Fixing ℓ_{mem} and ℓ_{cov} a priori protects against outcome-driven selection, but it invites the obvious question of whether the chosen depths are the right ones. We therefore repeat both axis validations at *every* one of the twelve RadioDINO-s16 blocks ($N = 150$, seed 42; Fig. 7). Features are extracted once per condition and reused across depths, so the sweep costs little more than the original experiments. The outcome does

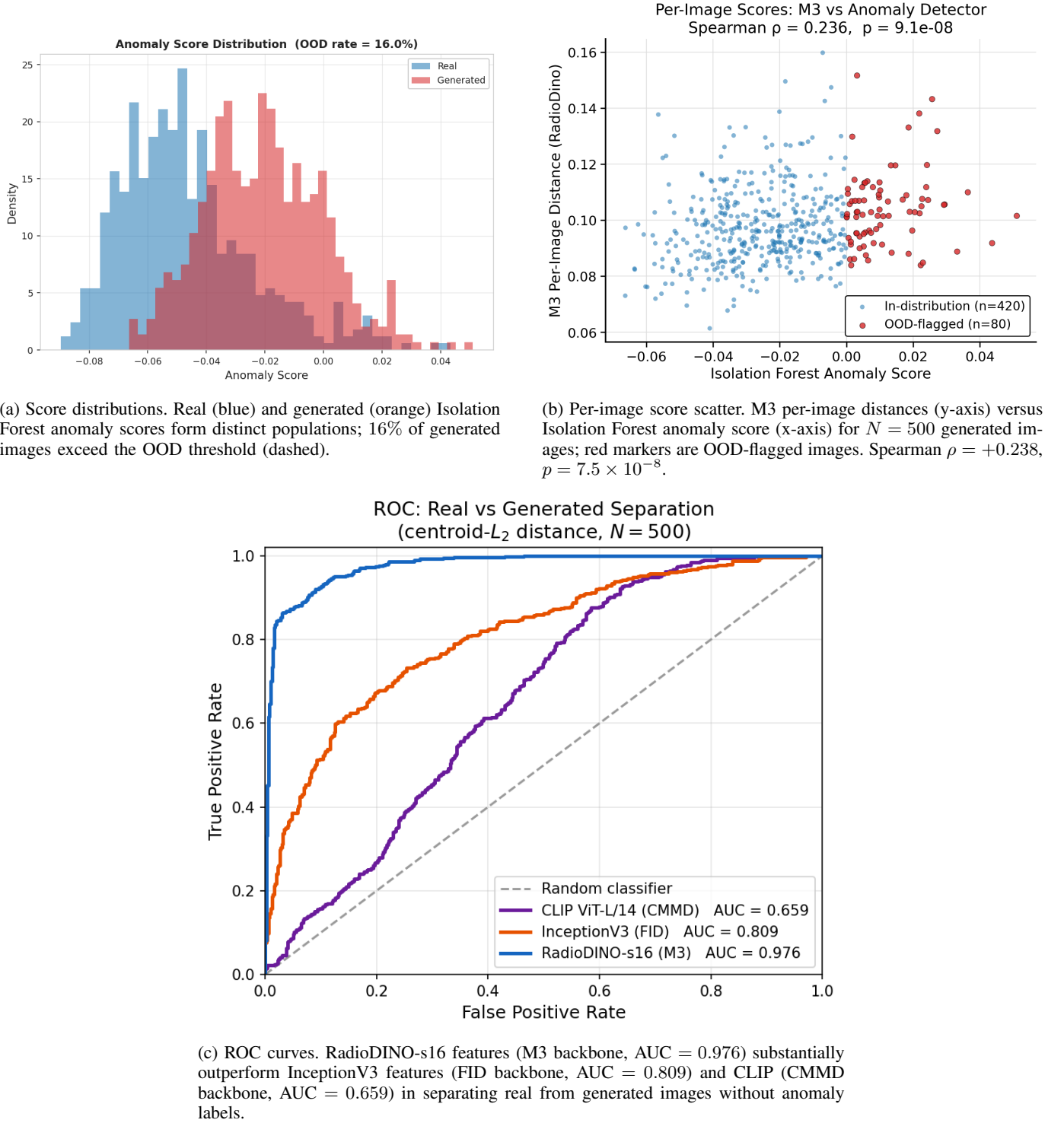


Fig. 6: Unsupervised anomaly detection results ($N = 500$ per class). (a) Score distributions show clear separation between real and generated populations. (b) M3 per-image scores correlate significantly with the independent Isolation Forest detector. (c) RadioDINO-s16 features achieve ROC-AUC 0.976, a 20.6% improvement over InceptionV3 (0.809) and a 48% improvement over CLIP (0.659).

not support promoting either prior to an empirically optimal choice, and we report it as such.

a) Memorization depth: Under the canonical protocol of Section IV-C (near-duplicates formed by jitter at $\sigma = 5 \times 10^{-4}$), the memorization rate recovers the injected fraction

exactly at all twelve depths: mean absolute recovery error is 0.000 everywhere. The test is saturated, so it certifies that the L9 prior costs nothing while providing no evidence that L9 is preferable to any other block.

To make the comparison discriminative we increase the

TABLE VIII: Memorization-axis recovery error (mean $|\text{recovered} - \text{injected}|$ over injection rates $\{0, 0.1, 0.2, 0.3, 0.5\}$) by block depth and near-duplicate difficulty σ . 0.220 is the error of a detector that flags nothing. The canonical protocol ($\sigma = 5 \times 10^{-4}$) is saturated; where the test does separate depths, the a-priori L9 is already past the ceiling.

σ	L1	L4	L9	L12	blocks with error < 0.01
5×10^{-4}	0.000	0.000	0.000	0.000	all 12
5×10^{-3}	0.000	0.000	0.000	0.000	all 12
0.02	0.000	0.000	0.220	0.220	L1–L5
0.05	0.000	0.220	0.220	0.220	L1
0.10	0.219	0.220	0.220	0.220	none

difficulty of the near-duplicates (Table VIII). A detection ceiling then appears and moves toward the input as σ grows: at $\sigma = 0.02$ blocks L1–L5 still recover the injection rate exactly while L6 and deeper fail outright (error 0.220, i.e. they flag nothing); at $\sigma = 0.05$ only L1 survives; by $\sigma \geq 0.1$ no depth detects anything. *Whenever the test separates depths at all, L9 is on the failing side of the ceiling.*

This result must be read with its construction in mind. The perturbation is additive pixel noise, which is precisely the kind of variation a deep self-supervised representation is trained to discard, so the protocol is biased toward shallow, near-pixel features by design. It therefore does not establish that a shallow block is the better memorization layer for the failure that matters clinically — a generator reproducing the *anatomy* of a training case. Adjudicating that requires near-duplicates perturbed semantically (small affine or elastic warps, intensity shifts) rather than by noise, which we did not run. What the sweep does establish is narrower and worth stating plainly: the L9 prior is not empirically corroborated, only shown to be harmless under the protocol on which the axis was originally validated.

b) Coverage depth: The mode-drop probe gives a cleaner answer. Recall should fall as generated diversity is removed; instead it *rises* at every one of the twelve depths, with Spearman ρ between drop fraction and recall positive throughout (L4: $\rho = +0.975$; the least-bad block, L8, still gives $\rho = +0.224$; six blocks give $\rho \geq +0.9$). No depth has the correct sign. The non-monotonicity reported in Section IV-C is therefore a property of the k -NN precision/recall estimator at this sample size, not a consequence of reading it at L4: changing the layer cannot repair it. This strengthens rather than softens our earlier caveat, and is the reason the coverage axis is reported as a diagnostic and excluded from every headline claim in this paper.

E. Generation-Quality Ladder

A metric must order images by known generation quality. We construct a quality ladder over the number of DDPM denoising steps $T \in \{10, 20, 50, 100, 1000\}$ (more steps = higher quality) from a fixed checkpoint, and compute the fidelity axis, FID, and CMMD against the real set. All three metrics are monotone in the correct direction (Table IX): each decreases as T rises. Most of the discriminative signal lies

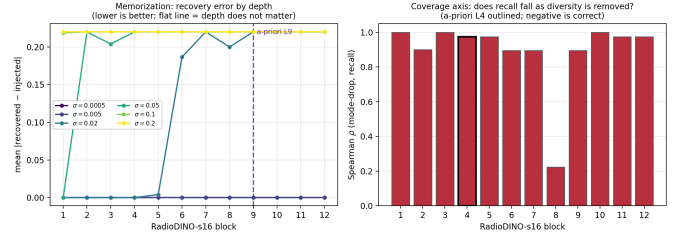


Fig. 7: Depth ablation over all twelve RadioDINO-s16 blocks ($N = 150$). **Left:** memorization recovery error against depth, one curve per near-duplicate difficulty σ . The two lowest σ curves lie flat at zero (every depth succeeds, so the test cannot rank depths); as σ grows a detection ceiling appears and retreats toward the input, and the a-priori L9 (dashed) is above it in every discriminative case. **Right:** Spearman ρ between mode-drop fraction and k -NN recall. Negative would be correct; every block is positive, so no depth fixes the coverage axis.

TABLE IX: Generation-quality ladder over DDPM denoising steps T ($N = 150$). All metrics decrease monotonically as quality improves (Spearman $\rho = -1.0$ for each).

Denoising steps T	M3 (fid.)	FID	CMMD
10	0.616	358.8	0.469
20	0.614	357.5	0.467
50	0.608	353.8	0.462
100	0.598	347.6	0.452
1000	0.187	132.4	0.112

between $T = 100$ and $T = 1000$; the metrics barely separate $T = 10$ from $T = 100$, a limitation of the noise-proxy construction we used to generate the ladder under compute constraints. M3 holds no unique advantage in ordering here. Its advantage is the native significance test (below), not monotonicity, which FID and KID also satisfy.

F. Lesion-Specificity with Real Expert Masks

A radiology-trained backbone should, in principle, respond more strongly to degradation of the diagnostically relevant region than to equal-area degradation of healthy tissue, a property natural-image backbones need not possess. We test this directly using the *real* expert FLAIR-abnormality segmentations of the lower-grade-glioma (LGG) dataset of Buda et al. [38] (600 image-mask pairs), replacing the intensity-threshold proxy masks of earlier drafts.

The test is a within-image, area-matched contrast. For each image we perturb (i) the annotated tumour region [LESION] versus (ii) an equal-area, equal-shape region relocated to healthy tissue [CONTROL]. We use two independent control constructions: a *mirror* control (the tumour mask reflected to the contralateral hemisphere) and a *texture-matched* control (the tumour mask translated to the healthy-brain location whose local gradient energy best matches the tumour's). Cases where a control lands off brain or on tumour are discarded, so lesion and control areas are identical by construction. We report the specificity ratio $\rho = \Delta_{\text{LESION}}/\Delta_{\text{CONTROL}}$, where Δ is the distributional shift (real vs. perturbed) induced by the

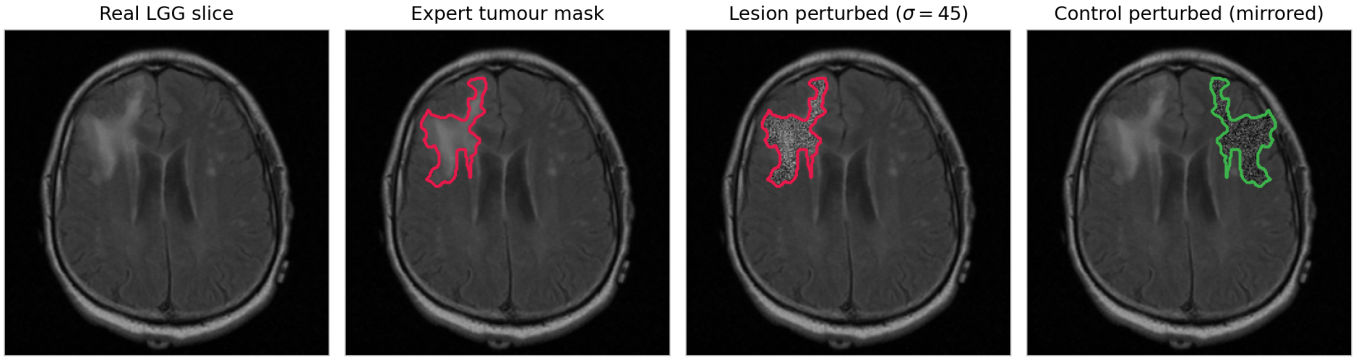


Fig. 8: The area-matched lesion-specificity test on a real LGG slice. From left: the raw slice; the expert FLAIR-abnormality mask (red); the LESION condition (region-local Gaussian noise, $\sigma = 45$, inside the tumour, red); and the CONTROL condition (the same noise inside the mirrored, equal-area contralateral region, green). Lesion and control regions are identical in shape and area, so any difference in metric response is attributable to *where* the perturbation is applied, not to how much of the image is changed.

perturbation; $\rho > 1$ means the metric reacts more to tumour damage than to equal-area healthy damage. The baseline is KID (unbiased Inception-MMD), which is valid at small N and lies in the *same estimator family* as M3, so the backbone is the only difference between the two. Figure 8 illustrates the two conditions.

Two regimes emerge. Under *gross erasure* (zeroing the region) no metric is lesion-specific ($\rho \approx 1.1$ – 1.3 for all): the black-hole signal dominates and swamps any semantic difference, consistent with, and explaining, the artefact-invariance of M3 discussed below. Under *subtle* region-local Gaussian noise, however, M3 responds more strongly when the perturbation falls inside the tumour than when the same perturbation falls in healthy tissue, whereas the Inception baselines respond almost equally to both. Figure 9 traces ρ against noise strength σ : in the measurable regime ($\sigma \geq 45$; below this the L12 MMD² underflows to zero at $N = 200$ and the ratio is undefined) M3 attains $\rho = 1.4$ – 1.8 against *both* controls, which agree to within 0.1, whereas the Inception baselines stay flat at $\rho \approx 1.03$ – 1.10 .

a) *Isolating the backbone from the kernel*: Attributing this gap to the *backbone* requires holding the estimator fixed, since M3 uses a multi-bandwidth RBF kernel while KID uses a degree-3 polynomial one; a comparison against KID alone varies both. We therefore add a kernel-controlled baseline: the *identical* multi-bandwidth RBF MMD² of Algorithm 3, applied to InceptionV3 features instead of RadioDINO ones, so that the backbone is the only difference from M3. It behaves like KID, not like M3 (Table X): across $\sigma \in \{45, 60, 90\}$ the RBF-on-Inception ratio is 1.03–1.10 and the polynomial-kernel KID ratio is 1.03–1.08, agreeing to within 0.04 at every σ , while M3 attains 1.39–1.81. Exchanging the kernel therefore accounts for essentially none of the gap, and the selectivity is attributable to the radiology backbone rather than to the estimator, to pixel magnitude, or to texture complexity — the last two being already controlled by the area-matched design and the 4% gradient-energy match of the mirror control.

Two caveats bound the claim. The effect decays as σ grows (extrapolating toward the $\rho \approx 1.1$ of gross erasure), so it is

TABLE X: Lesion/healthy specificity ratio ρ with bootstrap 95% CIs, mirror control, $N = 200$. The kernel-controlled baseline (RBF on InceptionV3) differs from M3 *only* in the backbone; KID additionally differs in the kernel. Both Inception baselines behave alike, so the separation from M3 is backbone-driven, not kernel-driven. “n/a” marks σ below a metric’s detection floor.

σ	M3 (RadioDINO, RBF)	Inception, RBF	Inception, poly. (KID)
20	n/a	1.13 [0.85, 1.32]	1.12 [0.85, 1.34]
30	n/a	1.09 [0.88, 1.26]	1.08 [0.88, 1.25]
45	1.71 [0.94, 2.14]	1.10 [0.92, 1.26]	1.08 [0.92, 1.24]
60	1.57 [1.03, 1.97]	1.10 [0.95, 1.24]	1.08 [0.93, 1.22]
90	1.39 [1.06, 1.71]	1.08 [0.96, 1.20]	1.06 [0.93, 1.19]

scoped as selectivity to *subtle* diagnostic-region degradation, not a universal pathology-sensitivity advantage. And at $\sigma = 45$ the M3 bootstrap interval still admits $\rho = 1$ ([0.94, 2.14] mirror); only at $\sigma = 60$ and 90 does the lower bound exceed unity (1.03 and 1.06 mirror; 1.02 and 1.03 texture-matched), so the individually significant evidence sits in that upper band rather than across the whole sweep. As no LGG generator was available, this is a metric-sensitivity test on real images under controlled perturbations, not a generator-quality result.

A second experiment isolates what each metric is actually responding to when a region is masked. Across three conditions ($N = 200$; Table XI) with a 64×64 center square mask on BraTS, masking the tumour region in real images (condition B) moves them *closer* to the tumour-free generated distribution in RadioDINO space (the M3 fidelity axis falls by 54.4%), whereas FID *increases* by 101.4% and CMMD by 191.7%.

The direction of these changes identifies the cause. If FID and CMMD were responding to the removal of the tumour, masking it in real images would move those images *towards* the tumour-free generated set and the distances would fall, as M3’s does. They instead rise sharply, and they rise by a similar amount in condition C, where the same square is applied to

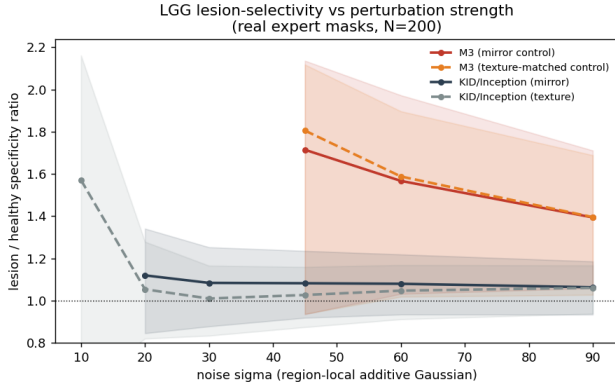


Fig. 9: Lesion/healthy specificity ratio ρ versus region-local noise σ on real LGG expert masks ($N = 200$). M3 (red/orange) is lesion-selective ($\rho = 1.4\text{--}1.8$) against both the mirror and texture-matched controls, which nearly coincide; the Inception-MMD baseline (grey) stays at $\rho \approx 1.05$. Bands are bootstrap 95% CIs. Points below the M3 detection floor ($\sigma < 45$) are omitted.

TABLE XI: Masking with a 64×64 center square mask ($N = 200$). B masks the center in real images; C masks the same region in the (tumour-free) generated images. Relative change is against the condition-A baseline. FID and CMMD rise steeply in *both* B and C, including in C where no tumour is removed, which identifies the black square rather than the anatomical change as what they are measuring.

Condition	M3	FID	CMMD
A: real vs. gen (baseline)	0.173	123.0	0.151
B: real vs. masked-real	0.079	247.7	0.439
C: real vs. masked-gen	0.215	267.7	0.493
Rel. change (B–A)/A	–54.4%	+101.4%	+191.7%
Rel. change (C–A)/A	+24.3%	+117.6%	+226.5%

generated images that contain no tumour to remove (+117.6% FID, +226.5% CMMD). Both metrics therefore respond to the black square itself rather than to the anatomical change, and the magnitude of the increase is not evidence of pathology sensitivity. M3 is comparatively invariant to the artefact and moves in the direction the anatomy implies. Read together with the LGG result, this is a statement about *specificity*: RadioDINO features track clinically meaningful change rather than the pixel-level artefacts that inflate FID and CMMD. The BraTS masks here are intensity-threshold FLAIR proxies, not ground truth, so this is reported as a supporting result only; the LGG experiment above, which uses real expert masks, carries the primary claim.

G. Sample Efficiency and Stability

We measure the coefficient of variation (CV) of each metric across random subsamples at $N \in \{25, 50, 100, 200, 500\}$ (5 repeats each). Table XII reports the result, and it does not favor M3 at small N : FID is more sample-stable than M3 for $N < 100$ (e.g. CV 3.2% vs. 17.0% at $N = 25$). When tested on fully independent subsets of $N = 500$,

TABLE XII: Coefficient of variation (%) across repeated draws; lowest per row in bold. FID is clearly the most stable metric at $N \leq 50$; M3 overtakes it from $N = 100$, and all three converge to $\sim 2\%$ by $N = 500$. Rows $N \leq 200$ use 5 random subsamples from the 500-image evaluation pool. The $N = 500$ row *cannot* be estimated this way — a 500-image subsample of a 500-image pool is the pool itself, giving a degenerate CV of exactly 0 — so it is instead estimated from 10 *independently drawn* 500-image evaluation sets.

N	M3 CV (%)	FID CV (%)	CMMD CV (%)	draws
25	16.98	3.18	15.09	5 subsamples
50	12.27	4.91	10.98	5 subsamples
100	2.36	2.87	6.75	5 subsamples
200	2.37	2.97	5.14	5 subsamples
500	1.93	1.89	1.82	10 independent sets

TABLE XIII: CKA threshold ablation on BraTS ($N = 500$, CLS-token CKA). S_{RG} : real-versus-generated equal-weight M3 score. S_{RR} : real-versus-real score on a held-out split. Disc. = $(S_{RG} - S_{RR})/S_{RR}$. Best raw discriminability: $\tau = 0.70$ (bold). Recommended default: $\tau = 0.80$ (underlined), balancing discriminability with multi-scale coverage.

τ	M	Active layers	Disc.
0.70	1	{12}	9.539
0.80	2	{1, 12}	9.423
0.85	2	{1, 12}	9.423
0.90	2	{1, 12}	9.423
0.92	4	{1, 2, 3, 12}	8.476
0.95	5	{1, 2, 3, 5, 12}	8.348
0.97	6	{1, 2, 3, 4, 5, 12}	8.053
0.99	8	{1–5, 7, 8, 12}	8.775

both metrics are highly stable: M3 attains CV 1.93% and FID attains 1.89%. M3 and FID converge to comparable CV ($\sim 2\text{--}3\%$) by $N = 100$, and the unbiased MMD² estimator underlying M3 (shared with CMMD) carries higher small-sample variance than FID’s Gaussian-moment estimator. We therefore recommend reporting M3 only for $N \geq 100$, and preferably $N = 500$.

H. CKA Threshold Ablation

This ablation is a *post-hoc check*, not a layer-selection mechanism: the canonical fidelity axis is fixed at L12 a priori (Section III-B). We include it only to verify that an adaptive CKA procedure, given a free choice, would not have preferred a different layer for separability. Table XIII reports discriminability $(S_{RG} - S_{RR})/S_{RR}$ across eight CKA thresholds. Features are extracted once for all images and the CKA selection is re-applied per threshold, so backbone loading cost is incurred only once. The deepest block (L12) is retained at *every* threshold, confirming that the a-priori fidelity layer is the one stable, always-selected representation regardless of the redundancy cutoff.

Across every threshold the most aggressive pruning ($\tau = 0.70$) retains exactly {12}, and the deep block L12 is never dropped. The discriminability curve is non-monotone in τ , but the qualitative conclusion is invariant: separability is carried by the deepest block, which is precisely the a-priori fidelity layer.

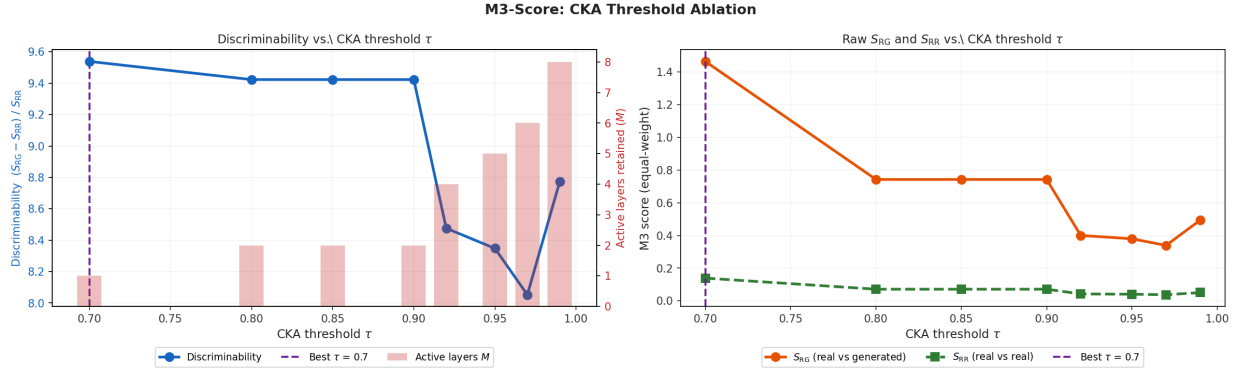


Fig. 10: CKA threshold ablation ($N = 500$, CLS-token CKA). *Left*: discriminability, defined as $(S_{RG} - S_{RR})/S_{RR}$ (blue, left axis), and the number of active layers M (red bars, right axis) versus threshold τ . The curve is non-monotonic: peak discriminability at $\tau = 0.70$ ($M = 1$, discriminability = 9.539) uses only L12, while $\tau = 0.80$ ($M = 2$, discriminability = 9.423) adds L1. This confirms that real-versus-generated separation is carried by the final layer.

A single-layer L12 fidelity score is therefore not a compromise but the configuration the data would have selected. This ablation does not feed back into the metric definition; it merely corroborates the structural choice of Section III-B (Fig. 10).

I. Statistical Significance: Permutation Test

We construct a null distribution by pooling the real and generated L12 fidelity features and permuting the labels 500 times, recomputing the unbiased MMD^2 on each shuffle (Section III-B). This is the native fidelity-axis permutation test, not an auxiliary construction.

The observed fidelity MMD^2 is 0.153. No permuted statistic out of $B = 500$ reached it. Using the standard add-one estimator, which avoids reporting an impossible $p = 0$, this gives $\hat{p} = (0 + 1)/(B + 1) = 1/501 = 0.002$. This is the smallest value the test can return at $B = 500$: it is a resolution floor, not evidence of a smaller p , and we therefore write $\hat{p} = 0.002$ rather than $p < 0.002$. A smaller reportable p would require more permutations, not a larger effect.

We do *not* treat the bootstrap interval’s exclusion of zero as evidence. The unbiased MMD^2 estimator is clamped at zero (Algorithm 3) and the population quantity is non-negative by construction, so a positive point estimate almost necessarily yields a CI above zero; the interval is reported to convey the *precision* of the estimate ($[0.146, 0.162]$, a width of 0.016), not as a test against the null. For the same reason we relegate the null-standardized separation Z : at $N = 500$ the observed value lies 446 null standard deviations above the null mean ($\sigma_0 = 3.4 \times 10^{-4}$), but this magnitude carries no inferential content. It grows approximately linearly with N by construction, because the null variance shrinks as $O(1/N)$ while the observed statistic converges to a positive constant, and the permutation null is strongly non-Gaussian, so Z maps to no tail probability. It is reported only as a descriptive signal-to-noise ratio and should not be compared across sample sizes. Inference rests on the permutation p -value and the bootstrap interval, both of which establish separability between the real and generated distributions well beyond conventional

thresholds and, unlike FID/KID/CMMD, are produced natively by the metric.

V. DISCUSSION

a) Why RadioDINO and not InceptionV3 or CLIP?:

Backbone alignment with the target domain is not a stylistic choice; it is a prerequisite for sensitivity. InceptionV3 is trained on 1000 natural-image classes and CLIP on web image-caption pairs. Neither has been exposed to the systematic co-variation of tissue intensity, pathological contrast, and acquisition artifact that defines a radiology distribution. The clearest evidence is the CMMD rank inversion (Table VI): CLIP features place the cross-modality CT shift *closer* to real brain MRI than a weak in-domain MRI generator ($\Delta_{CT-MRI} = -0.039$, 95% CI $[-0.059, -0.021]$, excluding zero), whereas the radiology-trained RadioDINO-s16 fidelity axis orders them correctly. The same backbone choice yields the OOD-AUC margin (0.976 vs. 0.809 for InceptionV3 and 0.659 for CLIP).

We stress that this is not a defect of the MMD estimator but a *misalignment of the CLIP embedding space with radiology modality structure*. CLIP is trained to align images with free-text captions, so its geometry is organized around high-level, modality-agnostic semantics (“a medical scan of a brain/chest”) rather than the Hounsfield-unit, tissue-texture, and acquisition statistics that govern pixel-level radiology fidelity. Under that objective, a lung CT and a brain MRI can be nearer in caption-semantic space than a brain MRI is to a *degraded* brain MRI; the CLIP embedding is doing exactly what it was trained to do. We therefore do not claim CLIP fails at natural-image evaluation, nor that CMMD is incorrectly implemented; we show that CLIP’s text-alignment ontology is structurally unfit for *ordering* radiology fidelity, which is precisely the gap a radiology-pretrained encoder closes.

b) *On the a-priori layers and the discarded adaptive design*: An earlier version of this work selected layers adaptively by CKA pruning and a discriminability criterion. We deliberately abandoned that design: selecting a layer because it maximises measured separability is selection on the test

outcome, which inflates apparent performance. The canonical metric now fixes L12/L9/L4 by representational position alone. The CKA ablation (Section IV-H) is retained only to show that this a-priori choice coincides with what an adaptive procedure would have picked for the fidelity axis (L12 is retained at every threshold), so we lose nothing by fixing it in advance and we gain immunity to outcome-driven selection.

c) *Where M3 does not win:* Three limitations remain. First, *sample efficiency*: FID is more stable than M3 for $N < 100$ (Table XII); the unbiased MMD² estimator pays for its lack of a Gaussian assumption with higher small-sample variance, so M3 should be reported at $N \geq 100$. Second, the *coverage axis* is not yet a validated claim: under mode-drop it behaves non-monotonically because k -NN radii inflate as the generated set shrinks (Section IV-C). Third, the *lesion-specificity* of M3 is real but narrow: with real expert masks it vanishes under gross erasure and appears only under subtle degradation ($1.4\text{--}1.8\times$ vs. an Inception baseline's $\sim 1.05\times$; Section IV-F). It is therefore artifact *specificity* to subtle, semantically meaningful change, not the raw-sensitivity advantage claimed in early drafts. A study with a lesion-conditioned generator is required to extend this from a metric-sensitivity test to a generator-quality claim. Fourth, the *layer priors for two of the three axes are unconfirmed*: the depth ablation of Section IV-D shows that L9 is indistinguishable from every other block under the protocol on which the memorization axis was validated, and that no depth at all gives the coverage axis the correct sign. Only the fidelity layer L12 is independently corroborated. We regard fixing the depths in advance as the right default, since it removes an obvious route to outcome-driven tuning, but it should be understood as a discipline rather than as an optimum we have demonstrated.

d) *Limitations and scope:* M3-Score is validated primarily on brain MRI (BraTS), with cross-modality checks on CT and retinal fundus. It is scoped to the radiology (CT/MRI) domain: RadioDINO was trained on radiology, and our cross-modality ordering shows M3 values are not comparable across the radiology boundary (Section IV-B). Extending validation to chest X-ray and CT generators with ground-truth pathology masks is the principal direction for future work. The metric also requires inference through a 12-layer ViT-S/16, which adds latency compared to FID: on a single consumer GPU the fidelity MMD² computation itself takes ≈ 8 s for 500 images, and $\approx 1\text{--}2$ minutes including the forward pass through the ViT for feature extraction. Finally, while $N = 500$ is standard for metric evaluation, larger-scale ($N \geq 1000$) and multi-site validation of the fidelity axis remains future work. Because the canonical metric fixes its layers a priori, it introduces no layer-selection or weighting hyperparameter; the only sample-size guidance is to use $N \geq 100$, since the unbiased MMD² estimator carries higher variance at small N (Table XII) and can return small negative values for $N < 50$.

e) *Connection to existing metrics:* The three M3 axes are complementary to manifold-based metrics such as Alpha-Precision and Beta-Recall. The fidelity axis measures *feature-level* shift with a domain-calibrated kernel and a native significance test; the memorization axis targets near-duplicate leakage; the coverage axis (with the caveats above) targets

diversity. We recommend reporting M3 alongside FID rather than in place of it: FID remains the more sample-stable scalar at small N , while M3 adds radiology-domain discrimination, the CMMD-style failure it avoids, and per-axis diagnostics.

VI. CONCLUSION

We introduced the M3-Score, a radiology-domain evaluation framework for generative image models built on the frozen RadioDINO-s16 ViT. Rather than a single tuned scalar, M3 reports two axes read from layers fixed a priori by representational depth: a fidelity axis (unbiased multi-bandwidth RBF MMD² at L12) with a native permutation p -value and bootstrap CI, and a memorization axis (L9) validated to recover injected near-duplicate rates exactly. A third, exploratory coverage diagnostic at L4 is specified and evaluated, but it fails its own mode-drop validation at all twelve encoder depths (Section IV-D); we therefore exclude it from every claim made here and report it only as a negative result, since the failure lies in the k -NN precision/recall estimator at these sample sizes rather than in the choice of layer. On BraTS brain MRI the fidelity axis separates real from generated images at permutation $\hat{p} = 0.002$, the resolution floor at $B = 500$ (bootstrap 95% CI $[0.146, 0.162]$) and detects out-of-distribution radiology at ROC-AUC = 0.976, versus 0.809 for InceptionV3 and 0.659 for CLIP. Our central result is the structural failure of CMMD, which *rank-inverts* a cross-modality CT shift and a weak in-domain generator, placing CT closer to real MRI (paired $\Delta = -0.039$, 95% CI $[-0.059, -0.021]$, excluding zero), where M3 (and FID) order them correctly.

Results that do not favor M3 are reported in full: FID is more sample-stable at $N < 100$; the coverage axis is unstable under k -NN radius inflation; and, with real expert lesion masks, M3 shows no lesion-specific advantage under gross erasure and only a modest one ($1.4\text{--}1.8\times$) under subtle degradation: artifact specificity, not raw sensitivity. The open-source implementation and a-priori layer configuration are released for reproducibility. Future work will extend validation to chest X-ray and CT generators, evaluate a lesion-conditioned generator against these real masks, and place the coverage axis on a firmer footing at larger sample sizes.

APPENDIX A

LEGACY ADAPTIVE AGGREGATION AND CKA LAYER SELECTION

This appendix specifies the one component of the superseded adaptive design that the reported results still depend on: the CKA-based layer selection used for the post-hoc validation of Section IV-H, which is what establishes L12 as the single most discriminative non-redundant layer. It is *not* part of the canonical M3-Score (Algorithm 1, Section III-B): the canonical axes read fixed layers and require none of this machinery, so no layer, threshold, or weight is selected on test outcomes.

The remainder of the abandoned design — the adaptive aggregation loop and the entropy-stability-uniqueness layer weighting — is not reproduced here, since no result in this

paper uses it. Its full specification and reference implementation are available in the released code repository (Section VI).

A. Post-Hoc Validation: CKA-Based Layer Analysis

a) *Motivation:* Adjacent transformer layers in DINOv2-style encoders frequently learn representations of high mutual similarity, particularly in the early and final blocks [39]. Including such layers would introduce correlated and computationally redundant contributions to the final score. The CKA [39] is used to identify a non-redundant subset $\mathcal{L}$ of the available layers L .

b) *Linear CKA:* Given two embedding matrices $E^{(\ell_1)}, E^{(\ell_2)} \in \mathbb{R}^{N_r \times D}$ extracted from a reference set of N_r real images, linear CKA is

$$\text{CKA}(E^{(\ell_1)}, E^{(\ell_2)}) = \frac{\|\tilde{E}^{(\ell_1)\top} \tilde{E}^{(\ell_2)}\|_F^2}{\|\tilde{E}^{(\ell_1)\top} \tilde{E}^{(\ell_1)}\|_F^2 \cdot \|\tilde{E}^{(\ell_2)\top} \tilde{E}^{(\ell_2)}\|_F^2}^{1/2}, \quad (11)$$

where $\tilde{E}^{(\ell)} = E^{(\ell)} - \mathbf{1}\bar{e}^{(\ell)\top}$ is the column-centered matrix, $\bar{e}^{(\ell)} = \frac{1}{N_r} \sum_i E_{i,:}^{(\ell)}$ is the column mean, and $\|\cdot\|_F$ denotes the Frobenius norm. $\text{CKA}(E^{(\ell_1)}, E^{(\ell_2)}) \in [0, 1]$, with value 1 indicating identical representational geometry.

Evaluating Eq. (11) directly requires forming $(D \times D)$ covariance matrices, whose denominator scales as $\|\cdot\|_F^4 \sim D^4$ and is numerically unstable. Following the HSIC reformulation in [39], we instead compute CKA via $(N_r \times N_r)$ gram matrices, which are independent of feature dimension:

$$K^{(\ell)} = \tilde{E}^{(\ell)} \tilde{E}^{(\ell)\top} \in \mathbb{R}^{N_r \times N_r}, \quad (12)$$

$$\text{CKA}(\ell_1, \ell_2) = \frac{\langle K^{(\ell_1)}, K^{(\ell_2)} \rangle_F}{\|K^{(\ell_1)}\|_F \cdot \|K^{(\ell_2)}\|_F}, \quad (13)$$

where $\langle A, B \rangle_F = \sum_{ij} A_{ij} B_{ij}$ is the Frobenius inner product, proportional to the Hilbert-Schmidt Independence Criterion.

c) *Greedy backward selection:* Given $\tau \in (0, 1)$ (default $\tau = 0.80$), a greedy pass iterating from layer $L - 1$ to layer 1 adds a layer to $\mathcal{L}$ only if its CKA similarity to every layer already in $\mathcal{L}$ is below τ . Layer L is always retained as the anchor. The full procedure is stated as Algorithm 5.

On BraTS brain MRI with $\tau = 0.80$, the procedure selects $M = 3$ layers $\mathcal{L} = \{1, 3, 12\}$, spanning early, mid, and deep transformer blocks. The number M of retained layers is data-adaptive and varies with both the imaging modality and the CKA threshold τ ; it is reported alongside S for reproducibility. In the BraTS evaluation, layers 1, 3, and 12 are retained because their pairwise CKA similarities all fall below $\tau = 0.80$, indicating that each captures a distinct aspect of the radiology-specific feature hierarchy [26], [33]: early layers encode local texture and edge responses, mid-network layers encode tissue boundaries and coarse anatomy, and the final layer encodes semantic pathology structure. CKA layer selection is performed once per evaluation session on $N_r = 20$ images using CLS token features and the resulting $\mathcal{L}$ is reused for the full evaluation pass.

Algorithm 5 CKA-based greedy layer selection (CKAPRUNE)

Require: Encoder Φ , reference set $\mathcal{X}_r^{(N_r)}$, threshold τ
Ensure: Active layer set $\mathcal{L} \subseteq \{1, \dots, L\}$

```

1: for  $\ell = 1, \dots, L$  do
2:    $E^{(\ell)} \leftarrow \text{EXTRACTFEATURES}(\Phi, \mathcal{X}_r^{(N_r)}, \{\ell\})$ 
3:    $\tilde{E}^{(\ell)} \leftarrow E^{(\ell)} - \mathbf{1}\bar{e}^{(\ell)\top}$   $\triangleright$  column-center, Eq. (11)
4:    $K^{(\ell)} \leftarrow \tilde{E}^{(\ell)} \tilde{E}^{(\ell)\top}$   $\triangleright (N_r \times N_r)$  gram matrix, Eq. (12)
5: end for
6:  $\mathcal{L} \leftarrow \{L\}$   $\triangleright$  layer  $L$  is retained unconditionally
7: for  $\ell = L - 1, L - 2, \dots, 1$  do
8:    $\text{redundant} \leftarrow \text{false}$ 
9:   for each  $\ell' \in \mathcal{L}$  do
10:     $\rho \leftarrow \langle K^{(\ell)}, K^{(\ell')} \rangle_F / (\|K^{(\ell)}\|_F \cdot \|K^{(\ell')}\|_F)$   $\triangleright$  Eq. (13)
11:    if  $\rho > \tau$  then
12:       $\text{redundant} \leftarrow \text{true}$ 
13:    break
14:   end if
15: end for
16: if  $\neg \text{redundant}$  then
17:    $\mathcal{L} \leftarrow \mathcal{L} \cup \{\ell\}$ 
18: end if
19: end for
20: return  $\mathcal{L}$ 

```

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